Tracing model-generated DNA with position-independent watermarking

Kimon Antonios Provatas[1,*] Christos Galanopoulos[1,*]
Ilias Georgakopoulos-Soares[1,†]

[1]The University of Texas at Austin, Austin, Texas, USA
[*]These authors contributed equally. [†]Correspondence: ilias@austin.utexas.edu

Abstract. Genomic language models can write synthetic DNA that carries no record of its origin. A generation-time watermark could provide a provenance signal, if a verifier can detect it using only the DNA sequence, a key, and published detector settings, without knowing where the generated region starts, which strand it lies on, or how it was divided into six-base tokens. We embed the SynthID tournament watermark in two genomic language models, Carbon and GENERator-v2. The verifier searches both strands, every start position, and four window lengths, and corrects its decision for the whole search. In a detection cohort of 1,544 held-out prompts, it found all 3,088 marked sequences per model, unedited and after one substituted, inserted, or deleted base. For ordinary sequences, the one-sided 95% upper confidence bound on the false-positive rate was at most 0.850%; one bound for the wrong-key control reached 1.015% (Carbon, after a deletion). In a development cohort, neither model showed a detectable change in likelihood or in predefined sequence measures. Under random edits placed without reference to the detector, detection stayed complete or nearly complete up to a 2% per-base edit rate. The detection and development cohorts share one layout, a 384-base prompt followed by a 3,072-base generated region. In a third, local cohort of 512 prompts generated on a different machine by the same authors, 54 layouts per model varied the region length from 384 to 3,072 bases, the natural DNA around the region, the strand, the token phase, and cropping, and at least 1,023 of 1,024 marked reads were detected in every layout. Under 64 keys never used in generation, no calibration fraction exceeded its nominal level on model output or on genomic DNA that no model generated. Regions shorter than 384 bases were not tested. These results show statistical detection, not biological function, secret-key security, or robustness to an editor who sees the detector.

1 Introduction

Genomic language models now generate coding sequences, regulatory elements, genome-scale constructs, and functionally constrained DNA designs [31, 38, 23]. A DNA sequence does not record whether it came from an organism, a design process, or a generative model, and that provenance is lost once the sequence leaves the system that produced it. One biosecurity proposal for designed proteins is to store synthetic gene sequences and synthesis data in repositories that are queried only in emergencies [3]. A provenance signal carried by the sequence itself would complement such records, because it cannot be separated from the sequence.

Watermarking embeds a statistical signal during generation, and a verifier holding the key can later test for it without the model, the prompt, or the generation logs [24]. In DNA, the verifier must first find where to look. Many genomic models write DNA as non-overlapping six-base tokens, but the emitted sequence does not show where one token ends and the next begins. We call the offset of that division the token phase. There are six phases, and the term is unrelated to alignment against a reference genome. Reverse complementation changes the tokens and their order, so both strands must be considered. When the generated region sits inside a longer sequence, its start and end are also unknown. The verifier has to search all of these possibilities and still control the probability of a false positive for the sequence as a whole.

We integrate the SynthID tournament watermark [9] into two genomic language models, Carbon-500M [4, 21] and GENERator-v2 1.2B [27, 12]. Both write six-base tokens and can sample directly from the same vocabulary. They are not independent, because the Carbon training corpus is built mainly on GENERator-v2's pre-training data [4]. SynthID preserves the model's output distribution in expectation over fresh keyed functions, and at a fixed key and context it samples from a deliberately reweighted distribution. We ask whether the sampler follows that distribution, whether watermarking changes likelihood or simple sequence statistics, and whether a verifier can find the mark without knowing the start, strand, or token phase.

The study uses three cohorts of prompts with separate roles. A development cohort of 256 prompts serves the sampler check, the quality comparison, and the choice of how to score a window. A detection cohort of 1,608 prompts, of which 1,544 are held out, serves detection, the false-positive rate, and the edit experiments. A third, local cohort re-generates 512 of those held-out prompts on a different machine and serves the layout, detector-comparison, calibration, and entropy studies. It is not an independent replication.

In both models, the verifier found every marked sequence in the detection cohort, unedited and after one edited base. For ordinary sequences, every one-sided 95% upper confidence bound on the false-positive rate lay below the 1% target. Neither model showed a detectable change in the quality measures. Random edits placed

without reference to the detector left detection complete or nearly complete up to a 2% per-base edit rate. After an insertion or deletion, the windows the verifier selected are consistent with regions where the token phase is preserved or recovered. In the local cohort, at least 1,023 of 1,024 marked reads were detected in each of 54 layouts per model, and calibration under 64 unused keys showed no fraction above its nominal level.

2 Background

2.1 Watermarking designed DNA

DNA watermarks predate generative models. The first synthetic bacterial genome carried watermark sequences [13], and a short signature was written into the genome of an engineered *Bacillus subtilis* strain [2]. DNA-Crypt hid messages in synonymous codons [18], and messages placed in an open reading frame without changing the encoded protein persisted during viral replication [28]. These designs need a location for the payload and attention to the biological effect of each change. Generation-time watermarking addresses a complementary setting by embedding the signal through the model's sampling distribution during generation, so it needs no payload region.

2.2 Watermarking language-model output

Language-model watermarks change how tokens are sampled. Some bias a pseudorandom subset of tokens and detect its excess frequency [24]. Others leave the output distribution unchanged in expectation over fresh keyed randomness [7, 26, 1, 20, 39]. SynthID in its non-distortionary configuration belongs to the second group. We apply it to DNA and propose no new construction. Robustness depends on what the attacker can do, and prior work describes evasion, key-recovery, and forgery attacks by stronger and adaptive attackers [40, 11, 36, 22, 35]. This study tests none of them; its edits are random and never consult the detector.

Text watermarks built to survive editing treat synchronization as a local problem. Kuditipudi et al. align a candidate text to a key sequence under an edit distance [26], and Kirchenbauer et al. search all contiguous spans and calibrate the false-positive rate for that search [25]. Both rely on an edit disturbing tokenization only near the edit. Golowich and Moitra tolerate a constant fraction of adversarial edits, but with a new construction built on pseudorandom codes [14]. A fixed vocabulary partition keeps the key synchronized under a positional shift, which roughly doubles the provable edit tolerance relative to the soft watermark keyed on the previous token without making the mark immune to edits [42]. Watermarks on sentence embeddings avoid tokenization but need an encoder and a natural segmentation into units [19], which a DNA verifier lacks.

Six-base tokens remove the locality these methods rely on. One inserted or deleted base shifts the reading frame, so every later token changes identity, and so does the keyed context that sets each mark bit. Alignment cannot repair this, because no token sequence survives to be aligned, and a fixed partition does not help, because the changed tokens carry independent mark bits. What survives is a contiguous region read in one of six phases. Verification therefore becomes a search over strands, phases, start positions, and window lengths, and the correction for that search is part of the method.

Four properties separate these constructions here: whether verification needs the generating model; whether it needs an alignment, a known boundary, or a known strand; which edits it claims to tolerate; and whether sampling is distorted. Biased-logit schemes distort sampling and are usually verified with known boundaries [24]. Distribution-preserving schemes realign to a key sequence [26] or certify a radius of token changes [39]. Watermarks for biological sequence assume a coding region [41] or mark the protein, not its DNA [5]. The construction evaluated here needs no model, alignment, known boundary, or coding region, and it preserves the sampling distribution in expectation.

2.3 SynthID

We use the binary tournament construction of SynthID-Text [9]. At each generation step, a secret key and the most recent tokens assign every candidate token a binary mark bit in each tournament layer, and each layer favors candidates whose mark bit is 1. Averaged over fresh keyed functions, the tournament leaves the next-token distribution unchanged. At a fixed key and context, it samples from a reweighted distribution. The detector rebuilds the mark bits from the output and tests for an excess of 1s. Because the bits depend on local context, not on absolute position, they can be rebuilt from any retained stretch of sequence, and the detector needs only the output, the key, and published detector settings (Section 3.3). Keying on recent tokens is common to autoregressive text watermarks [24]. We chose SynthID because it combines local keying with tournament sampling and has been evaluated in a large-scale text deployment.

2.4 Genomic tokenization and token phase

Several genomic models, including both studied here, divide DNA into non-overlapping six-base tokens, which shortens the token sequence and lets a model cover more bases in a fixed context [8, 38, 4, 27]. Models with one token per base have no phase problem, because an insertion or deletion moves tokens without changing them [31, 32]. Caduceus uses one token per base because small input changes can otherwise change the tokenization drastically [37]. GENERator randomizes the tokenization offset of each training sample [38], and GENERator-v2 cycles through all six offsets [27]. None of these remedies is available to a verifier that receives only DNA, which must recover the phase from the sequence itself.

3 Methods

3.1 Models and sampling

Generation used Carbon-500M [4, 21] and GENERator-v2-eukaryote-1.2b-base [27, 12], two decoder-only models. At every step, we restricted the logits to the 4,096 canonical DNA tokens and normalized once. For GENERator, this is its direct token distribution, not the per-base sampling that its released code applies by default. Temperature was 1.0, with no top-k or top-p truncation. The resulting distribution p is shared by both samplers: the ordinary sampler draws from p without a key, and the marked sampler applies the tournament below. We report each model separately.

3.2 Watermark embedding

At each step, the key, the preceding four tokens, and a candidate token determine the candidate's mark bit g, computed with HMAC-SHA-256 over the key, a fixed domain label, the context, and the candidate. The tournament formula, the number of layers, the context width, and the repeated-context rule follow the public reference implementation of SynthID-Text [15]. One tournament layer reweights the distribution as

$$p'(x) \;=\; p(x)\,\Big(1 + g(x) - \textstyle\sum_y p(y)\,g(y)\Big), \qquad (1)$$

so candidates with a mark bit of 1 gain probability by an amount set by the total probability of such candidates. The sampler applies $m = 30$ layers, each with its own mark bits, and samples from the resulting distribution p_k (Figure 1a).

The first four generated tokens are sampled ordinarily, so every later context can be rebuilt from the continuation alone. A four-token context that has already occurred is also sampled ordinarily and excluded from the detector score, as in the original construction [9]; the sampler keeps a history of 1,024 contexts to recognize repeats. Averaged over fresh keyed functions, Equation 1 leaves the next-token distribution unchanged. At a fixed key and context, the sampler follows the reweighted distribution p_k, so the sampler checks test against p_k.

3.3 Position-independent detection

The verifier receives a DNA sequence, the key, and the published detector settings: the generation domain label, the four window lengths, the 30 tournament layers and four-token context used to rebuild the mark bits, the 1,024-context repetition history, and the significance level $\alpha = 0.01$. In deployment, only the key would be secret. It does not receive the prompt, the model or its probabilities, the seed, the strand, the token phase, or the generation boundary (Figure 1b).

The verifier treats every candidate generated region as a window: a stretch of the sequence on one strand, starting at a given base, with one of four lengths. It scores windows at every starting base of the sequence and of its reverse complement, which covers every start position and all six token phases. Inside a window, the DNA is divided into tokens from the window's first base, and the first four tokens supply context. Every later token whose context is not a repeat is a scored token. The verifier rebuilds the mark bits of the scored tokens and counts the ones, S, among their n mark bits.

If the sequence carries no mark for this key, the mark bits behave as independent fair coin flips, and the window P-value is the binomial tail

$$P_{\text{win}} \;=\; \Pr\big[\text{Binomial}(n, \tfrac{1}{2}) \geq S\big]. \qquad (2)$$

For a sequence of N bases, the verifier scores

$$M \;=\; 2\sum_{L}(N - L + 1), \qquad L \in \{384,\ 768,\ 1{,}536,\ 3{,}072\}, \qquad (3)$$

windows, which is 16,136 for a 3,456-base sequence, and makes one decision for the sequence:

$$\begin{aligned} P_{\text{seq}} \;&=\; \min\big(1,\ M \cdot \min_{\text{windows}} P_{\text{win}}\big), \\ \text{mark found} \;&\iff\; P_{\text{seq}} \leq \alpha. \end{aligned} \qquad (4)$$

The Bonferroni factor M covers every window searched, including overlapping ones, and the level α was set in advance. We call $-\log_{10} P_{\text{win}}$ the strength of a window. A sequence is positive exactly when some window reaches strength $\log_{10}(M/\alpha) = 6.21$. The search depends only on the sequence, so it is the same for both models, and M is recomputed when an edit changes the sequence length. Supplementary Section S4 gives the cost of the search.

Bonferroni correction needs no independence between tests, but it is conservative here, because windows of the same length whose starts differ by a multiple of six bases share a token phase and nearly all their tokens (Supplementary Section S3).

A window need not be intact to be detected. A token that has lost its phase or its context contributes unbiased bits and only dilutes the count, so a window is positive once it holds enough tokens that kept both. In the shortest window, about 56% of the mark bits must be 1 (Supplementary Section S2).

3.4 Prompts and cohorts

Prompts are 384-base forward-strand windows from 16 accession-versioned RefSeq chromosome records of eight model organisms. Windows were chosen by hashing a fixed label, the accession, and a segment index. Human, patient, controlled-access, organelle, plasmid, and ambiguous-base spans were excluded. Each cohort was fixed before any model generated from it, and no prompt was added, moved, or removed afterward.

The development cohort contains 256 prompts. Before any detection result was examined, a hash rule reserved 64 of them for choosing how to summarize the mark bits in a window (Supplementary Section S1) and held out the other 192. The sampler check and the sequence measures use this cohort only.

The detection cohort contains 1,608 prompts, 134 non-overlapping windows from each of 12 of the 16 records,

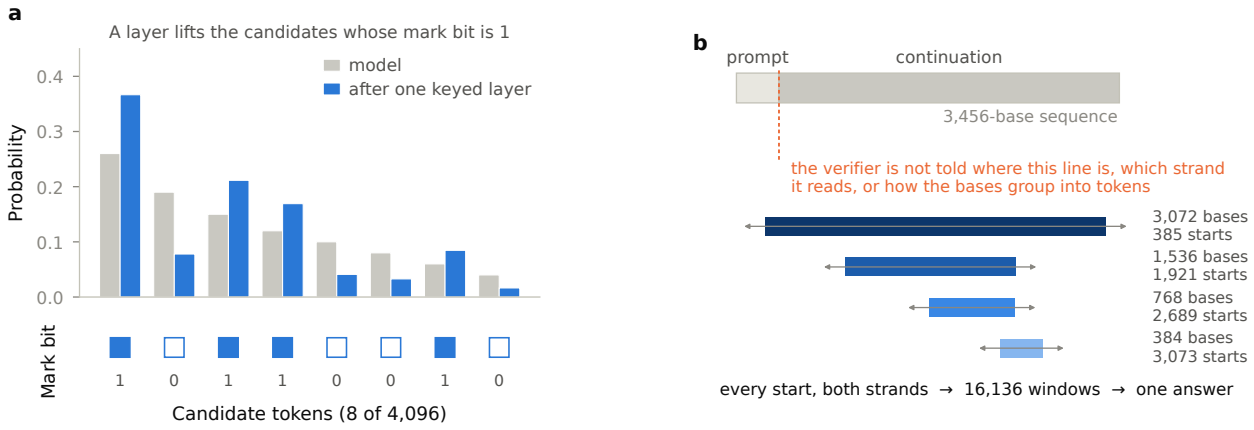


Figure 1: SynthID embedding and position-independent detection. **a**, One keyed layer, drawn for eight of the 4,096 candidate tokens with an illustrative distribution. Candidates whose mark bit is 1 gain probability, and the others lose it. The sampler applies 30 such layers. Panel **a** illustrates Equation 1 and is not a measurement. **b**, Detector inputs and search. The verifier receives the sequence, the key, and the published detector settings, without the generation boundary, the strand, or the token phase. It scores windows of four lengths at every starting base on both strands, 16,136 windows for a 3,456-base sequence, and reports one sequence-level result corrected for the full search.

covering six of the eight organisms (not mouse or budding yeast), drawn with a separate selection label. The same hash rule set aside 64 prompts, which were not used, and held out 1,544. A separate generation domain label gave both models new keyed functions, and all other settings matched. The same authors generated and scored this cohort with the same code, in a separate computing environment.

Every prompt produced two marked and two ordinary continuations of 512 tokens (3,072 bases) per model. The two draws used different seeds, and the two marked draws used different keys, both published for reproducibility. Prepending the prompt to a held-out continuation gives the 3,456-base test sequence that the verifier receives, either a marked sequence or an ordinary sequence; its generation boundary is withheld. In these two cohorts every test sequence therefore has the same standard layout: the generated region is its last 3,072 bases, lies on the forward strand, and starts a multiple of six bases from the first base. The local cohort varies this layout (Section 3.9). Scoring each marked sequence with the other marked draw's key gives the wrong-key control, which tests whether a positive depends on the correct key. The three rates are reported separately, because a pooled accuracy would depend on how common marked sequences are in an application.

3.5 Sampler validation

The correctness of the marked sampler rests on implementation tests in the project's test suite. The tournament update reproduces the public SynthID-Text reference implementation [15] on peaked, flat, random, and degenerate distributions. The vectorized computation of p_k used during generation agrees with an independent unvectorized implementation to floating-point precision. Computing p_k once and applying its inverse-CDF sampling step reproduces the generation sampler's draws exactly.

A second check uses realistic model states. For each model and development prompt, we fixed the state at token index 64 of the first ordinary continuation and drew 5,000 samples from each of p and p_k. Each set of counts was compared with its source distribution by the likelihood-ratio (G) statistic. An expected count of about 1.2 per category is too low for the asymptotic χ^2 reference, so G was referred to a Monte Carlo null of 999 samples from the same distribution, which is exact by construction. Because the draws come from the computed distributions, this check confirms the sampling step and the test, not the computation of p_k. The 256 tests per sampler form one family under Benjamini–Hochberg correction at 0.05. Bonferroni correction cannot reject here, because 999 replicates allow no P-value below 10^{-3}, above its level of 1.95×10^{-4}. As a negative control, at eight further states per model, draws from p were tested against p_k.

3.6 Sequence measures

The model score is the teacher-forced negative log-likelihood per generated token under the same model and token restriction as generation, with the prompt as context only. Each model scores its own outputs, so scores are not compared between models. Thirteen further measures were specified before analysis: GC content; purine content; CpG content; base, dinucleotide, and trinucleotide entropy; distinct 6-mer fraction; longest and mean single-base run; Jensen–Shannon drift from the prompt at $k = 1, 2, 3$; and perplexity. All 14 measures form one family under Benjamini–Hochberg correction.

3.7 Edits and statistics

In the single-edit conditions, each sequence has exactly one substituted, inserted, or deleted base in the continuation, placed by a fixed pseudorandom rule that never consults the verifier. Prompts are the independent unit, and a prompt's two draws are paired repetitions. Uncertainty in the sequence measures used an equal-weight bootstrap over prompts with 20,000 replicates and a paired sign-flip test with 100,000 replicates. A control prompt is positive if either draw is positive; for marked prompts, we also require both draws to be detected.

The detection cohort's analysis plan, specified before generation, reports the exact one-sided 95% upper confidence bound (Clopper–Pearson) on the false-positive rate over the 1,544 held-out prompts. Ordinary sequences, the false positives that matter in use, are the primary control, and the wrong-key control is a secondary check of key specificity. The prompt-level rate is about twice the per-sequence rate that α controls, so comparing it with the 1% target is conservative. Development-cohort rates carry exact two-sided 95% intervals.

3.8 Edit-rate experiment

To find where random editing defeats the verifier, we rescored the detection cohort's stored sequences after applying $k = \mathrm{round}(3{,}072\, r)$ edits to each continuation. We call r the per-base edit rate and tested seven values from 0.03% to 10% (Figure 5). A substitution changes one base. Each insertion or deletion adds or removes one to five bases, so the edit rate counts edit events, not changed bases. Four series were run: substitutions, insertions only, deletions only, and a mixed series of random insertions and deletions. The single-direction series keep opposite frame shifts from cancelling. Positions and indel lengths came from a public hash stream set by the series, the sequence, and the rate, never by the detector, and no sequence or detector setting changed. Marked sequences were scored with the correct key and ordinary sequences with the corresponding key, 3,088 sequences per model, rate, and series, without wrong-key controls. The analysis plan required the ordinary-sequence rate to stay at or below 1% in every cell, because a rise with edit rate would mean that the edits themselves create apparent signal.

3.9 Local cohort and layout studies

The detection and development cohorts cannot show how detection depends on the layout, whether the P-value is valid across keys, or what explains differences in strength. We therefore generated a third cohort on one Apple M5 Pro machine (PyTorch MPS backend, bfloat16) and ran four studies on it, with detection on the machine's CPU (Supplementary Section S4). The local cohort uses the first 512 held-out prompts of the detection cohort in its split order, with two marked and two ordinary continuations of 512 tokens per prompt and model. New generation labels gave both models new keyed functions and new sampling seeds, and all other settings matched the detection cohort. The same authors generated and scored this cohort with the same code. It repeats the detection cohort's design on a subset of its prompts on different hardware. It is not an independent replication, and we do not pool it with the detection cohort. With 512 prompts, a control with no positive prompt has an exact one-sided 95% upper confidence bound of 0.583%, and the bound is below 1% only if at most one prompt is positive. We call every sequence that the verifier receives a read, and the prompt is the independent unit throughout.

Standard layout. The unchanged verifier scored the local cohort at the standard layout, in the four edit conditions and with the three controls of the detection cohort.

Layout study. The layout study has 54 layouts per model, fixed before generation by public hash rules that never consult the detector, a key, or a score. The marked region of a read is a marked continuation or its first l bases. The control read carries the ordinary continuation of the same prompt and draw in the same layout, with the same flanks. A flank is natural genomic DNA, not model output: it is drawn from the 1,608 public source spans of the detection cohort (3,072 bases each, from the 12 RefSeq records that gave the prompts), and a flank longer than one span joins pieces from several spans, none of them the read's own. Six series vary one aspect each, except series C, which varies phase and strand. Series A truncates the region, with l from 384 to 3,072 bases. Three of the seven lengths (576, 1,152, and 2,304) fall between window lengths. Series B embeds the same regions in a 6,144-base read, with a natural left flank of 1,536 bases. Series C places a region of 768 or 3,072 bases after a left flank of $384+o$ bases, with o from 0 to 5, so that the region starts at each of the six token phases, and supplies each read forward or as its reverse complement. Series D surrounds a region of 768 or 3,072 bases with flanks of 0, 384, 1,536, or 4,608 bases on each side. Series E places a 768-base region at five positions in a 6,144-base read. Series F crops the standard read to 768, 1,536, or 2,304 contiguous bases at a hashed start. Two checks were specified before the run. Series A at 3,072 bases must reproduce the standard-layout decisions exactly, and a series C read and its reverse complement must give the same smallest window P-value and number of windows. A failure of either would be an implementation fault, not a result. No detection rate for marked reads had a pass threshold, and the ordinary rate, with the prompt as the unit, was expected at or below 1% in each layout.

Detector comparison. On the first 256 prompts, five settings were scored by four detectors, all with the same window statistic and the same 0.01 sequence-level target, each corrected for the windows it searches. The five settings are the standard layout, the standard layout after one insertion, and series B at 768, 1,152, and 3,072 bases. The aligned oracle is told the strand, the start of the generated region, and its length, and scores that one window; after the insertion it is told the original start and length, not the edit. The phase-only search scores the whole read from each of the six phases on both strands, 12 windows. The verifier is unchanged. The fine-grid search scores every start on both strands and every length $384+96j$ that fits the read. It stands in for a search over all contiguous

spans, which we did not run. These detectors are analysis baselines. No verifier setting changed, and the baselines are used for no other result. We summarize the margin of a read by its corrected strength, $-\log_{10}$ of the sequence-level P-value, which is at least 2 exactly when the read is positive at $\alpha = 0.01$. Cost is the median single-thread time of one detection call, measured on 32 prompts.

Calibration across keys. The verifier ran unchanged under 64 keys that were never used in generation, each derived from its index by hashing a public label. Three sources of reads were scored, 512 ordinary and 512 marked reads per model at the standard layout (first draw) and the 1,608 natural source spans, each searched with its own number of windows. We counted decisions at the 0.01 level and the fraction of decisions with a sequence-level P-value at or below 0.01, 0.05, 0.10, 0.25, and 0.50. The rule was set in advance: the P-value is supported as valid for a source if every observed fraction is at or below its nominal level, and a fraction whose exact one-sided 95% lower bound exceeds its nominal level is evidence against validity. Decisions on one read under different keys share the read, and decisions under one key share the key, so the exact intervals treat dependent decisions as independent. We also report the fraction of reads positive under at least one key.

Entropy. One teacher-forced pass of the generating model over each continuation recorded the Shannon entropy of the 4,096-token distribution at every generated position, averaged per sequence. We report the Spearman correlation of a marked sequence's mean entropy with its strength at the standard layout and with the fraction of its mark bits equal to 1 in the full continuation window. Intervals are 95% percentile intervals from a bootstrap over prompts with 10,000 replicates. A positive correlation was expected before the measurement.

4 Results

4.1 Sampler fidelity

At the tested model states, draws from the fixed-key and model distributions agreed with those distributions, and the test could tell the two distributions apart. Each model had 256 tested states, and chance alone predicts 12.8 of them to reject at $P < 0.05$. The marked sampler produced 13 in Carbon and 5 in GENERator, and the ordinary sampler produced 12 and 14. In GENERator, neither sampler had a rejection after Benjamini–Hochberg correction. For Carbon, the corrected count was not retained with the evidence, and its nominal counts are close to chance. The negative control, draws from the model's distribution tested against the fixed-key distribution, was rejected at all eight control states in each model. Together with the implementation tests (Section 3.5), these results support the correctness of the sampler at the resolution of 5,000 draws per state. They do not prove it correct at every possible state.

4.2 Sequence-quality proxies

Watermarking produced no statistically detectable difference between marked and ordinary continuations in either model (Figure 2). Every interval for the 14 measures contained zero, and no comparison was significant after correction. The smallest adjusted P-value was 0.85 in Carbon and 0.73 in GENERator. The paired difference in model score was +0.00852 nats per token in Carbon and −0.00564 in GENERator. The largest absolute standardized effects were 0.084 standard deviations in Carbon, on the longest single-base run, and 0.111 in GENERator, on the mean single-base run. These results indicate no measurable quality loss under the evaluated proxies.

4.3 Detection and false-positive rate

In the detection cohort, the verifier found every marked sequence in both models, unedited and after each single-base edit (Table 1; Figure 3). That is 3,088 sequences per model and condition, and it places the exact two-sided 95% interval for the prompt-level detection rate at 99.76–100%.

Every one-sided 95% upper confidence bound on the false-positive rate for ordinary sequences lay below the 1% target. The largest was 0.850%, for GENERator after a deletion. The analysis plan had stated before generation that a bound below 1% required at most eight positive prompts.

The wrong-key control, a secondary check of key specificity, crossed 1% in one cell. For Carbon after a deletion, its bound was 1.015%.

The correction was conservative in practice. Per sequence, the corrected decision was positive for 0.105% of ordinary Carbon sequences and 0.186% of ordinary GENERator sequences, pooled across the four conditions, well below α (Supplementary Section S3).

The development cohort gave the same result. All 384 marked sequences per model were found in every condition, and each control produced at most one positive sequence per model and condition (Supplementary Table S1).

Typical detection strength was the same in both models. The median unedited marked sequence reached strength 792.9 in Carbon and 799.7 in GENERator, far above the threshold. The weakest detected sequences differ more, but in the development cohort that difference came from a single GENERator sequence with a highly repetitive continuation (Supplementary Section S6).

4.4 Detection after single-base edits

One inserted or deleted base cost a sequence almost half of its strength, and one substituted base cost little, yet every marked sequence stayed detectable (Figure 4a,c). A substitution lowered the median strength by about 2%, and an insertion or deletion by about 45% (Supplementary Table S2).

An insertion or deletion also changed which window carried the evidence. For unedited and substituted sequences, the full 3,072-base window was strongest in

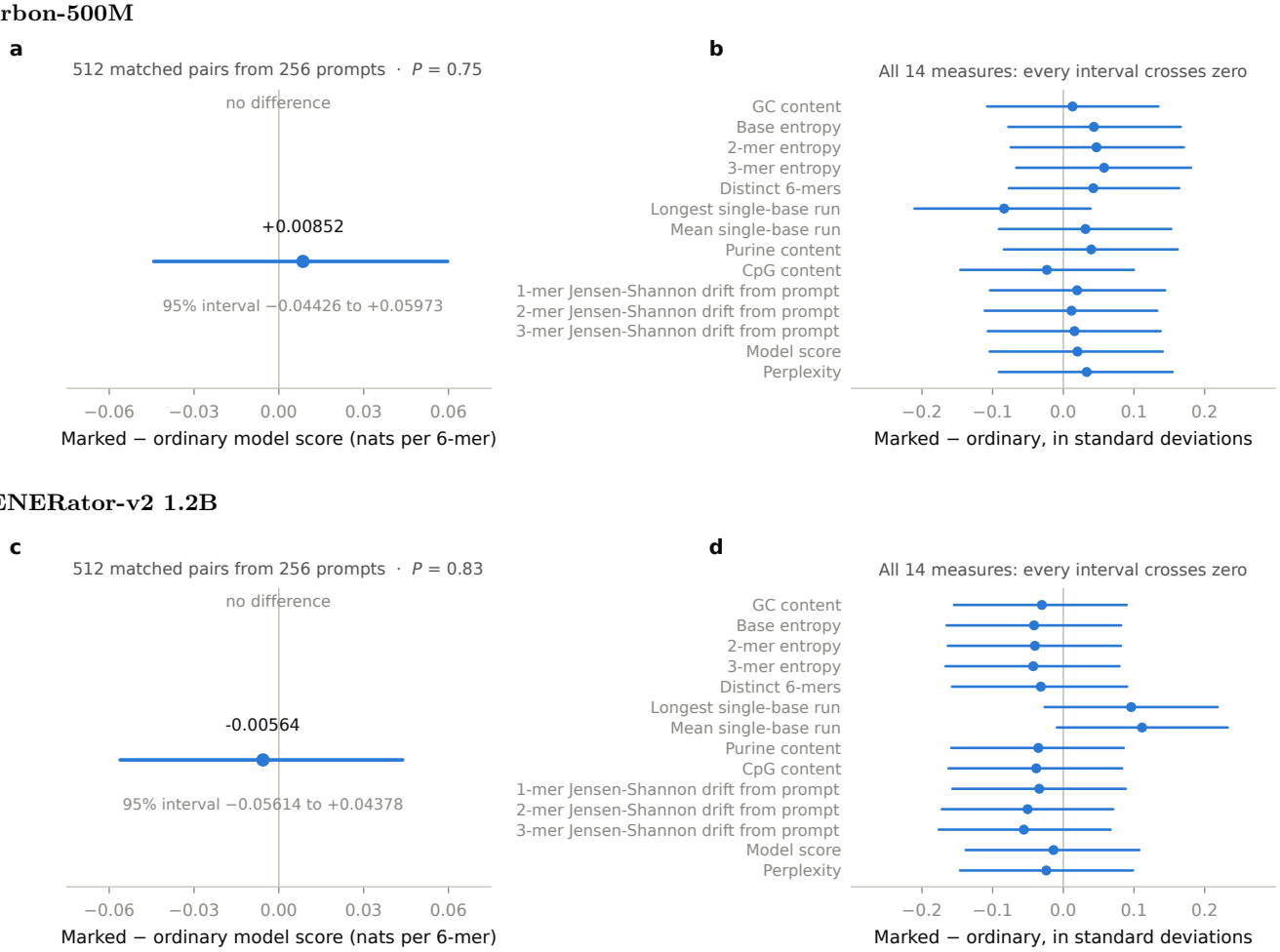


Figure 2: Model likelihood and sequence-quality proxies in the development cohort. a,c, Paired difference in model score between marked and ordinary continuations, over 512 matched pairs from 256 prompts. Positive values indicate lower model likelihood for marked continuations. The horizontal bar is the 95% interval from a bootstrap over prompts: −0.04426 to +0.05973 in Carbon ($P = 0.75$) and −0.05614 to +0.04378 in GENERator ($P = 0.83$). **b,d**, Paired differences for all 14 measures in standard deviations, with intervals in the same units. Every interval includes zero, and none of the 14 comparisons is significant after multiple-testing correction. Panels **a,b** show Carbon and panels **c,d** show GENERator.

Table 1: Positive detections in the detection cohort, for each model across 1,544 held-out prompts with two draws each (3,088 sequences per model and condition). Correct-key entries count marked sequences. Control entries count positive sequences, each from a different prompt; the value in parentheses is the exact one-sided 95% upper confidence bound on the prompt-level rate, in percent. Ordinary sequences were designated in advance as the primary false-positive measure, and the wrong-key control is a secondary check of key specificity. Each edit changes exactly one base.

	Carbon-500M			**GENERator-v2 1.2B**		
Edit applied to the sequence	Correct key	Ordinary	Wrong key	Correct key	Ordinary	Wrong key
None	3,088/3,088	3 (0.501)	5 (0.680)	3,088/3,088	5 (0.680)	2 (0.407)
One substitution	3,088/3,088	3 (0.501)	5 (0.680)	3,088/3,088	5 (0.680)	2 (0.407)
One insertion	3,088/3,088	3 (0.501)	6 (0.766)	3,088/3,088	6 (0.766)	2 (0.407)
One deletion	3,088/3,088	4 (0.592)	9 (1.015)	3,088/3,088	7 (0.850)	3 (0.501)

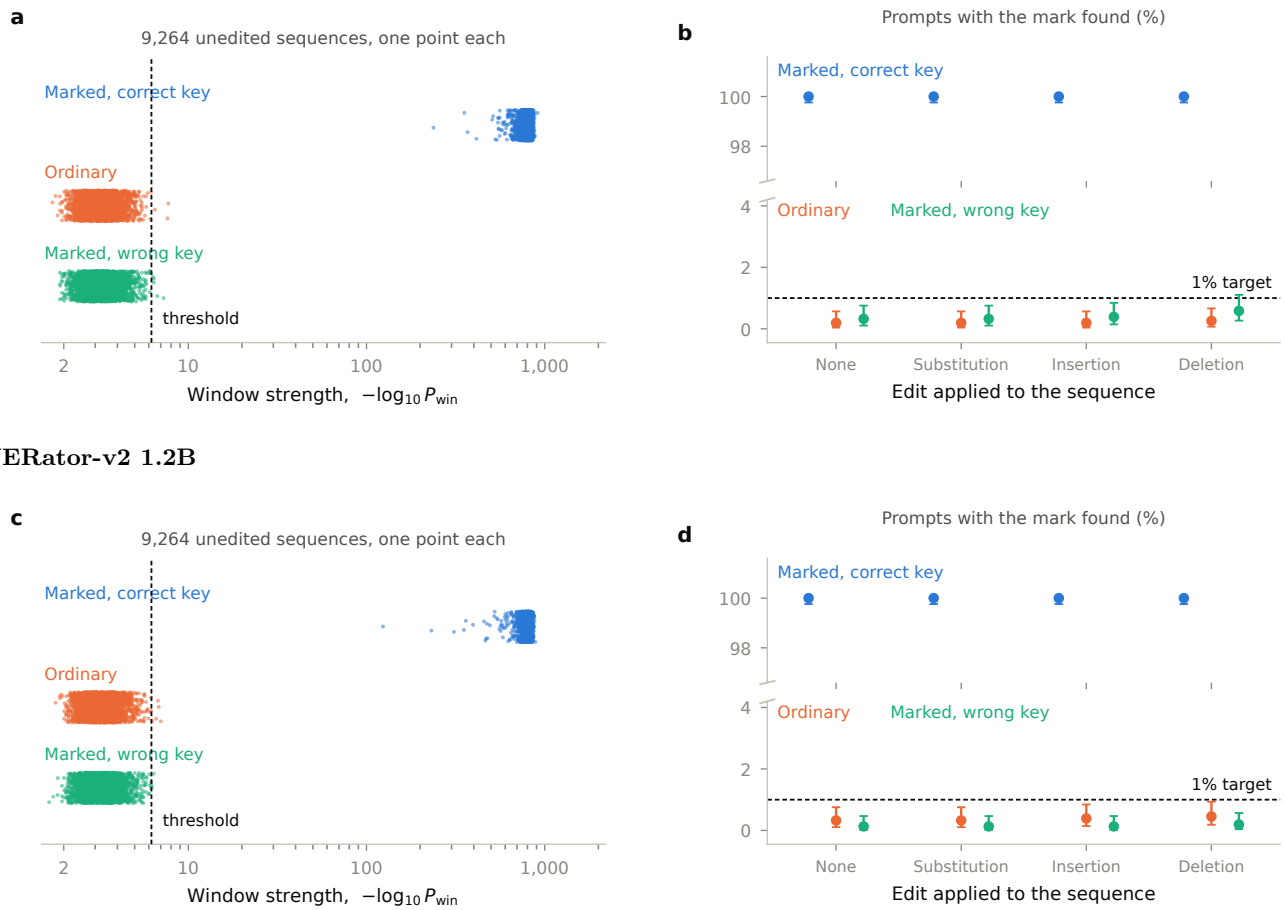


Figure 3: Position-independent detection in the detection cohort. a,c, Strength of the best window in each unedited sequence, one point per sequence, on a log scale: 3,088 marked sequences scored with the correct key, 3,088 ordinary sequences, and 3,088 wrong-key controls per model. The dashed line is the threshold corrected for the search over 16,136 windows. All marked sequences exceed it. **b,d**, Proportion of the 1,544 held-out prompts with positive detections (both draws for marked sequences with the correct key, either draw for controls), with exact two-sided 95% intervals, on a broken axis. All marked prompts were detected in all four edit conditions. Table 1 gives the counts and the one-sided upper bounds used for the false-positive comparison, and Supplementary Figure S1 shows the development cohort. Panels **a,b** show Carbon and panels **c,d** show GENERator.

nearly every sequence. After an insertion or deletion, a shorter window was strongest in almost half of the sequences (Figure 4b,d). A substitution changes token content and nearby contexts but keeps the downstream token phase. An insertion or deletion shifts the division of every later token in a window that spans the edit. The shift toward shorter windows is consistent with the search selecting regions whose token phase is preserved or recovered. The development cohort showed the same shift (Supplementary Figure S2).

4.5 Detection as edits accumulate

Detection stayed complete or nearly complete through a 2% per-base edit rate, that is, 61 edits per continuation, in both models (Figure 5). Every marked sequence was found at every rate up to 2% for every edit type, except for one GENERator sequence in the mixed series at 2%.

Insertions and deletions defeated the verifier at lower rates than substitutions. At a 5% rate, substitutions were still detected in every Carbon sequence and all but one GENERator sequence, whereas detection after insertions and deletions fell to 92.9–93.9%. At 10%, substitutions were detected in 19.7% of Carbon sequences and 18.7% of GENERator sequences. The other edit types fell to 4.6–5.9%.

The two models' curves agreed within one percentage point at every rate and edit type, so the single GENERator miss at 2% does not indicate a difference in robustness. The insertion-only and deletion-only series behaved like the mixed series, so cancelling frame shifts do not explain the tolerance.

Ordinary sequences stayed below the 1% target at every rate and in every series. Pooled over all rates and edit types, 0.15% of ordinary Carbon sequences and 0.13% of ordinary GENERator sequences were positive. The largest value in any single cell was 0.29%, for GENERator in the mixed series at 1%.

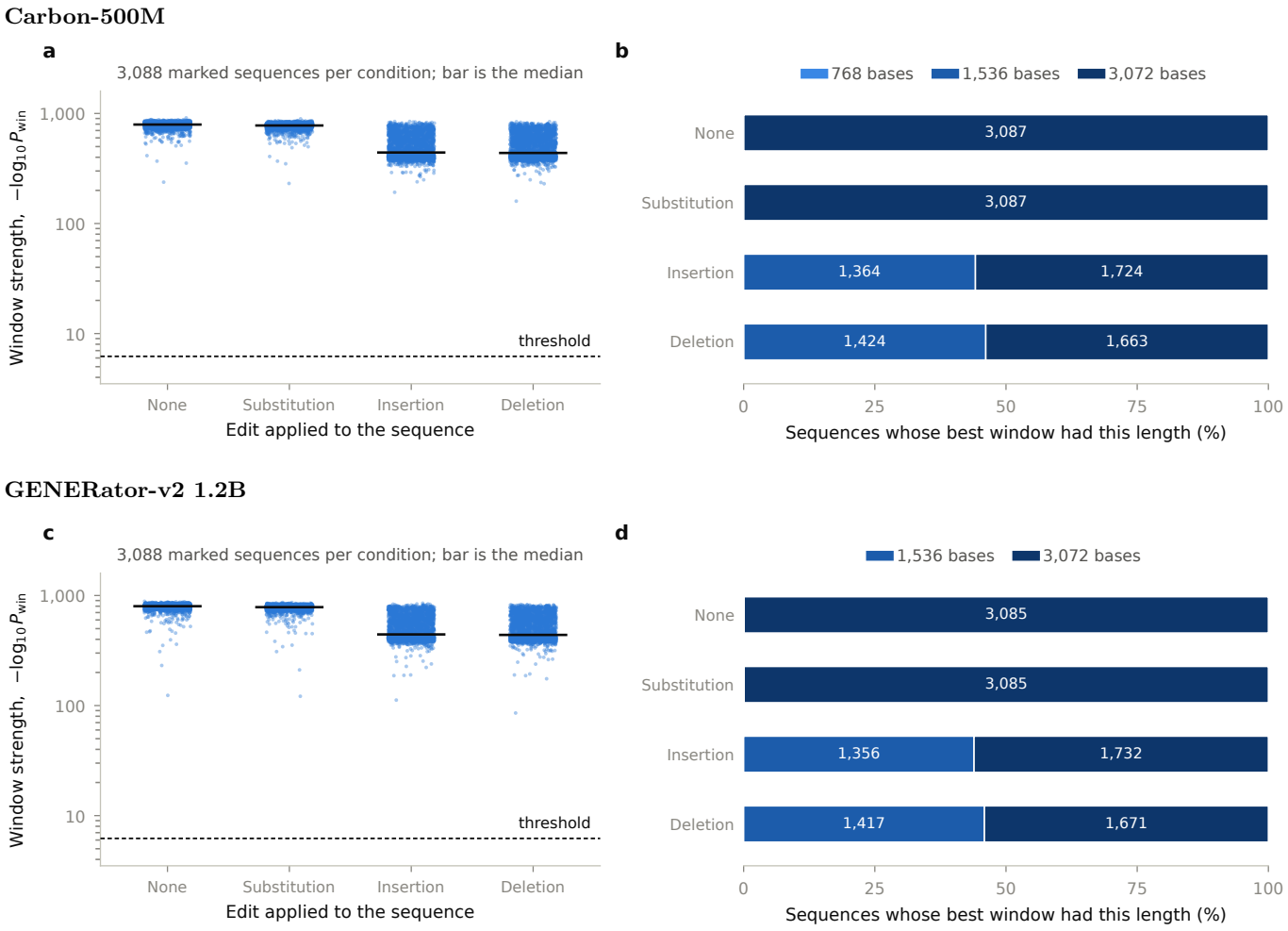


Figure 4: Effects of single-base edits on detection in the detection cohort. a,c, Strength of the best window in each of the 3,088 marked sequences per condition, on a log scale, with the median shown as a bar. The dashed line is the detection threshold. **b,d**, Length of the strongest window; numbers inside the bars are sequence counts out of 3,088. The full 3,072-base window is strongest for nearly all unedited and substituted sequences, whereas shorter windows are selected for almost half of the sequences after an insertion or deletion. Supplementary Figure S2 shows the development cohort. Panels **a,b** show Carbon and panels **c,d** show GENERator.

4.6 Detection in the local cohort

At the standard layout, the local cohort gave the same pattern as the detection cohort (Supplementary Table S3). The verifier found all 1,024 marked reads, with both draws detected for all 512 prompts, in every edit condition in both models. The median strength of unedited marked reads was 791.5 in Carbon and 797.4 in GENERator, close to the 792.9 and 799.7 of the detection cohort. One inserted or deleted base again lowered it by close to half, to 437.4–444.3 in Carbon and 437.5–450.9 in GENERator.

Ordinary reads were positive in 3, 2, 2, and 2 prompts in Carbon (unedited, substitution, insertion, deletion) and in none, none, none, and one prompt in GENERator. The wrong-key control was positive in one or two prompts in every condition in both models. At 512 prompts, the one-sided 95% upper bound is 0.583% for no positive prompt, 0.923% for one, 1.224% for two, and 1.507% for three. Every control with two or more positive prompts therefore has a bound above 1%. This cohort has about a third as many prompts as the detection cohort, whose 1,544 prompts remain the evidence for the false-positive bound.

4.7 Detection across layouts

Detection did not fail in any of the 54 layouts in either model. At least 1,023 of the 1,024 marked reads were detected in every layout (Table 2; Supplementary Table S4). Both checks specified before the run passed in both models. Series A at 3,072 bases reproduced all 2,048 standard-layout decisions without a mismatch, and all 24,576 forward and reverse-complement pairs in series C gave the same smallest window P-value and number of windows.

The misses are few. In Carbon, one marked read was missed, in one layout: a 384-base region between natural flanks. In GENERator, one marked read was missed in each of the 28 layouts whose region or crop is 768 bases or shorter. Every GENERator read was detected at 1,152 bases and longer. In both models the lowest rate

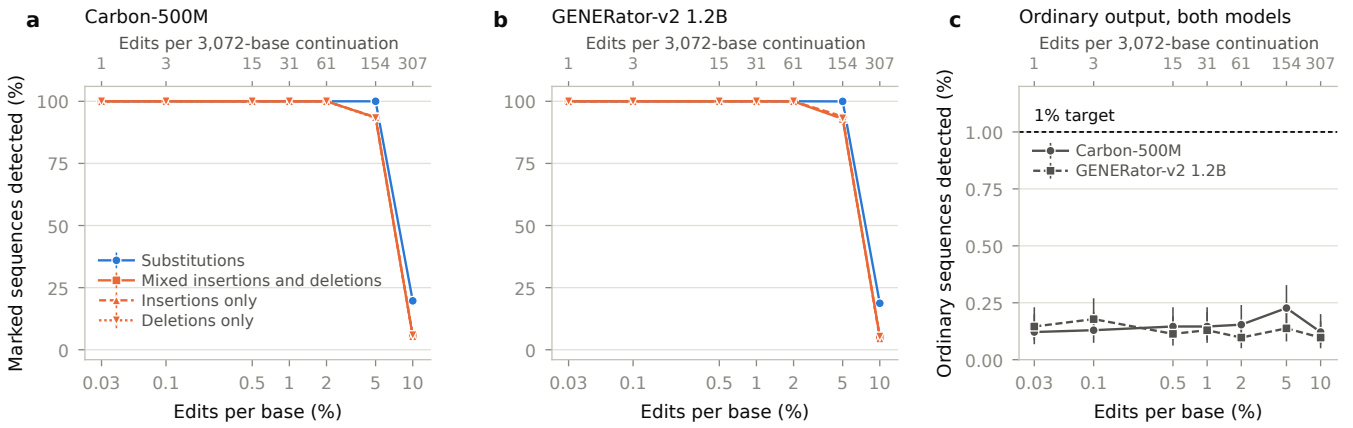


Figure 5: Detection as random edits accumulate, detection cohort. **a,b**, Percentage of 3,088 marked sequences detected with the correct key in Carbon (**a**) and GENERator (**b**), by per-base edit rate (log scale) and edit type. The rates were 0.03, 0.1, 0.5, 1, 2, 5, and 10%; the top axis gives the corresponding 1, 3, 15, 31, 61, 154, and 307 edits per 3,072-base continuation. Each insertion or deletion adds or removes one to five bases, and in the mixed series each edit is an insertion or a deletion at random. **c**, Percentage of ordinary sequences detected under the corresponding key, pooled over the four edit types (12,352 sequences per point), against the 1% target. Pooled over every rate and edit type, including the unedited sequences, 132 of 89,552 Carbon sequences and 116 of 89,552 GENERator sequences were positive. Vertical bars are exact two-sided 95% intervals and are often smaller than the markers. Edit positions were chosen without reference to the detector.

Table 2: Detection across layouts in the local cohort, for representative layouts. For each model, *Reads* are marked reads detected (of 1,024, from 512 prompts with two draws), *Prompts* are prompts with both draws detected (of 512), *Strength* is the median strength of the best window before the correction for the search, and *Ordinary* is the number of ordinary reads flagged (of 1,024). In series A the region follows the prompt, in series B it lies between natural flanks in a 6,144-base read, and in series D the region length is given before a plus sign and the flank on each side after it; in series F the read is a random crop of the standard read. Lengths are in bases. Regions of 576 and 1,152 bases fall between window lengths. Supplementary Table S4 lists all 54 layouts.

	Carbon-500M				**GENERator-v2 1.2B**			
Layout	Reads	Prompts	Strength	Ordinary	Reads	Prompts	Strength	Ordinary
A, 384	1,024	512	94.9	3	1,023	511	94.0	6
A, 576	1,024	512	106.8	3	1,023	511	106.5	5
A, 1,152	1,024	512	216.5	5	1,024	512	216.1	3
A, 3,072	1,024	512	791.5	3	1,024	512	797.4	0
B, 384	1,023	511	94.9	3	1,023	511	94.0	0
B, 576	1,024	512	110.8	2	1,023	511	110.4	3
B, 1,152	1,024	512	223.4	1	1,024	512	223.7	2
B, 3,072	1,024	512	791.5	1	1,024	512	797.4	1
D, 768 + 0	1,024	512	194.4	4	1,023	511	194.8	3
D, 768 + 4,608	1,024	512	194.6	1	1,023	511	194.8	0
D, 3,072 + 4,608	1,024	512	791.5	1	1,024	512	797.4	1
F, crop 768	1,024	512	105.2	3	1,023	511	105.7	5
F, crop 2,304	1,024	512	407.8	3	1,024	512	409.7	0

in any layout was 1,023 of 1,024 reads, which is 511 of 512 prompts with both draws detected.

Median strength tracked the length of the marked region and little else. At 384 bases it was 94.9 in Carbon and 94.0 in GENERator, and at 3,072 bases 791.5 and 797.4, about eight times as large for eight times the length. For regions of the four window lengths, embedding the region between natural flanks (series B), moving it (series E), lengthening the flanks to 4,608 bases on each side (series D), starting it at another token phase, or reverse-complementing the read (series C) left the median strength within 0.5 of the series A value. At 768 bases it ranged from 194.4 to 194.8 in both models, and at 3,072 bases it was 791.5 in Carbon and 797.4 in GENERator in every layout. Regions whose length falls between two window lengths were detected with a median strength close to that of the next shorter window length. At 1,152 bases it was 216.5 in Carbon and 216.1 in GENERator, against 194.6 and 194.8 at 768 bases, and the lengths of 576 and 2,304 bases showed the same pattern (Supplementary Table S4). Between natural flanks the medians at these lengths were higher: in Carbon, 223.4 against 216.5 at 1,152 bases and 446.7 against 408.7 at 2,304 bases. Random crops of 768, 1,536, and 2,304 bases were detected in every read except one GENERator read at 768 bases.

The ordinary rate was expected to stay at or below 1% in every layout. At read level it did. The largest count was 6 of 1,024 reads (0.586%) in GENERator, for a 384-base region, and 5 of 1,024 (0.488%) in Carbon. At prompt level, counting a prompt as positive if either draw is, one layout did not meet the expectation: the 384-base region in GENERator, with 6 of 512 prompts (1.172%) and a one-sided 95% upper bound of 2.30%. Its marked rate carries that caveat. In Carbon the largest prompt-level rate was 5 of 512 (0.977%). With 512 prompts the upper bound exceeds 1% in every layout with two or more positive prompts, which is most Carbon layouts and a minority of GENERator layouts (Supplementary Table S4). In those layouts these controls do not show a rate below 1%.

Detection was at its ceiling in every layout, so these data locate no boundary at which it fails within the tested range. They show that strength falls with the length of the region, not that detection does. Regions shorter than 384 bases, which is 64 tokens, were not tested, and the flanks come from the same 12 RefSeq records as the prompts.

4.8 Comparison with narrower and denser detectors

Where the layout is aligned, the verifier detected as many marked reads as an aligned oracle, and after an insertion it recovered detection that the oracle lost (Table 3). The comparison is descriptive. We call no detector better where the exact 95% intervals of the rates overlap. At the standard layout, all four detectors detected all 512 marked reads in both models. They did the same for a 3,072-base region in a 6,144-base read, and the oracle, the verifier, and the fine grid did so at 768 and 1,152 bases.

The search recovers detection after an insertion. The oracle, told the original start and length, detected 486 of 512 marked reads in Carbon (230 of 256 prompts with both draws) and 483 of 512 in GENERator (227 of 256) after one inserted base. The verifier, the phase-only search, and the fine grid detected all 512 in both models. These are the only intervals that do not overlap in the comparison: the oracle against each of the three searches, after the insertion, in both models. The oracle's median corrected strength fell from 791.5 to 197.2 in Carbon and from 798.3 to 202.5 in GENERator, whereas the verifier's fell from 787.3 to 451.2 and from 794.1 to 460.9.

For short regions in long reads, the search over starts and lengths gave a higher median corrected strength than the phase-only search. This comparison is descriptive. For a 768-base region in a 6,144-base read, the phase-only search, which scores the whole read, had a median corrected strength of 22.2 in Carbon and 22.9 in GENERator, against 189.6 and 190.7 for the verifier, and it missed one read in each model (511 of 512). At 1,152 bases the phase-only strengths were 51.8 and 52.9 against 219.4 for the verifier, and the phase-only search missed one Carbon read. At 3,072 bases they were 381.5 and 385.0 against 786.9 and 793.7.

The search buys no detections where the layout is aligned. There the verifier's median corrected strength (787.3 in Carbon, 794.1 in GENERator) was below the oracle's (791.5 and 798.3), the cost of the larger correction. A denser search raised the margin without adding detections. At 1,152 bases the fine grid reached 289.2 in Carbon and 290.3 in GENERator, against 219.4 for the verifier, and after the insertion 591.8 and 596.3, against 451.2 and 460.9. It took about five times as long per read at the standard layout (224.1 against 45.1 ms in Carbon, 224.6 against 44.4 ms in GENERator) and about eight times as long in the 6,144-base reads (741.9 against 96.7 ms and 746.2 against 97.0 ms at 1,152 bases). Single-thread time per read ranged from 0.48 to 5.04 ms for the oracle, 27.3 to 55.3 ms for the phase-only search, 41.5 to 97.0 ms for the verifier, and 221.8 to 746.2 ms for the fine grid. The fine grid stands in for a search over all contiguous spans, which was not run.

The baselines flagged more ordinary reads than the verifier, which corrects for more windows. At read level, the phase-only search flagged 6 of 512 ordinary reads at the standard layout in both models and 7 of 512 at 1,152 bases in GENERator. The oracle flagged up to 5 of 512 and the phase-only search up to 7, whereas the verifier flagged at most 2 and the fine grid at most 1. Counting a prompt as positive if either draw is, the oracle and the phase-only search exceeded 1% in several settings (3 to 7 of 256 prompts; Supplementary Table S5), and the verifier and the fine grid did not. At 256 prompts no control, including the verifier's, has a one-sided 95% upper bound below 1%.

Table 3: Four detectors on five settings in the local cohort, 256 prompts and 512 reads per model. *Windows* is the number of windows searched per read. *Marked* and *Ordinary* are the marked reads detected and the ordinary reads flagged, both of 512. *Strength* is the median corrected strength of marked reads, $-\log_{10}$ of the sequence-level P-value, and a read is positive at 2 or more. *ms* is the median single-thread time per read on the local machine; it is not comparable with the timing on a different processor in Supplementary Section S4. Supplementary Table S5 gives the prompt-level counts and bounds.

		Carbon-500M				**GENERator-v2 1.2B**			
Detector	Windows	Marked	Ordinary	Strength	ms	Marked	Ordinary	Strength	ms
Standard layout									
Aligned oracle	1	512	2	791.5	4.92	512	4	798.3	4.97
Phase only	12	512	6	696.4	28.8	512	6	701.3	29.0
Verifier	16,136	512	2	787.3	45.1	512	0	794.1	44.4
Fine grid	101,442	512	0	786.6	224.1	512	0	793.3	224.6
Standard layout, one insertion									
Aligned oracle	1	486	3	197.2	1.76	483	2	202.5	1.78
Phase only	12	512	3	397.8	27.3	512	2	403.7	27.5
Verifier	16,144	512	1	451.2	41.6	512	0	460.9	41.5
Fine grid	101,508	512	0	591.8	221.8	512	0	596.3	222.5
768-base region, embedded									
Aligned oracle	1	512	5	194.2	0.48	512	1	195.3	0.49
Phase only	12	511	2	22.2	54.8	511	5	22.9	55.3
Verifier	37,640	512	0	189.6	91.4	512	1	190.7	90.8
Fine grid	351,482	512	0	188.6	739.1	512	0	189.7	742.8
1,152-base region, embedded									
Aligned oracle	1	512	4	294.7	1.92	512	2	295.8	1.95
Phase only	12	511	5	51.8	50.6	512	7	52.9	51.1
Verifier	37,640	512	1	219.4	96.7	512	2	219.4	97.0
Fine grid	351,482	512	0	289.2	741.9	512	1	290.3	746.2
3,072-base region, embedded									
Aligned oracle	1	512	2	791.5	4.99	512	4	798.3	5.04
Phase only	12	512	2	381.5	49.7	512	4	385.0	50.1
Verifier	37,640	512	1	786.9	93.4	512	0	793.7	92.4
Fine grid	351,482	512	1	786.0	739.8	512	0	792.8	740.3

4.9 Calibration across keys

Under 64 keys that were never used in generation, the corrected decision was positive in at most 0.140% of decisions in every source, and no calibration fraction exceeded its nominal level (Table 4; Supplementary Table S6). Ordinary output was positive in 37 of 32,768 decisions in Carbon and 36 in GENERator, marked output scored with unused keys in 46 (Carbon) and 42 (GENERator), and natural genomic spans in 132 of 102,912. The positive rate of a single key ranged from 0 to 0.781% for ordinary output in both models, to 0.586% (Carbon) and 0.781% (GENERator) for marked output, and to 0.435% for natural spans, and no key reached 1%. At the other four levels, the observed fractions ranged from 0.525 to 0.638% at the 5% level, 0.983 to 1.147% at 10%, 2.521 to 2.719% at 25%, and 4.587 to 4.782% at 50%, about a tenth of the nominal level at each. By the rule set before the run, the P-value is supported as valid for these sources, and it is conservative: no one-sided lower bound exceeded its nominal level.

The decisions on one read under different keys share that read, and the decisions under one key share the key, so the pooled intervals treat dependent decisions as independent. The read-level view is therefore the more cautious summary. Of 512 ordinary reads, 36 were positive under at least one of the 64 keys in each model (7.03%), as were 45 of 512 marked reads in Carbon, 40 in GENERator, and 127 of 1,608 natural spans (7.90%). Those fractions are close to 64 times the per-decision rate. The result describes these sources, one derivation of the keys, and one sequence length (3,456-base reads and 3,072-base spans). It says nothing about secret-key security.

4.10 Entropy and strength

Marked sequences with a higher mean entropy had a higher strength in both models (Table 5). The Spearman correlation of a sequence's mean entropy with its strength was 0.663 in Carbon and 0.581 in GENERator, and with the fraction of its mark bits equal to 1 it was 0.653 and 0.580. Every bootstrap interval excludes zero. This is a correlation, not a causal attribution. Mean entropy was similar for marked and ordinary continuations and between the models: 7.595 and 7.598 nats in Carbon, and 7.563 and 7.563 in GENERator, against 8.318 for a uniform distribution over the 4,096 tokens. Each model

Table 4: Calibration of the sequence-level decision under 64 keys never used in generation, in the local cohort. *Positive / decisions* counts decisions with a corrected P-value of at most 0.01 (512 reads times 64 keys for model output, 1,608 spans times 64 keys for natural spans). *Per-key rate* is the range of the positive rate over keys. The last column counts reads positive under at least one key. The decisions share reads and keys, so the counts are not independent. Supplementary Table S6 gives the fractions at five nominal levels.

Source of reads	Positive / decisions	Rate (%)	Per-key rate (%)	Reads positive under ≥ 1 key
Carbon-500M, ordinary output	37/32,768	0.113	0–0.781	36/512 (7.03%)
GENERator-v2 1.2B, ordinary output	36/32,768	0.110	0–0.781	36/512 (7.03%)
Carbon-500M, marked output	46/32,768	0.140	0–0.586	45/512 (8.79%)
GENERator-v2 1.2B, marked output	42/32,768	0.128	0–0.781	40/512 (7.81%)
Natural genomic spans	132/102,912	0.128	0–0.435	127/1,608 (7.90%)

Table 5: Spearman correlation of the mean entropy of a marked sequence with its strength and with the fraction of its mark bits equal to 1, in the local cohort (1,024 sequences from 512 prompts per model). Brackets give 95% intervals from a bootstrap over prompts with 10,000 replicates.

Model	Strength	Mark bits equal to 1
Carbon-500M	0.663 [0.620, 0.703]	0.653 [0.610, 0.694]
GENERator-v2 1.2B	0.581 [0.532, 0.627]	0.580 [0.531, 0.626]

had a tail of low-entropy sequences. The lowest mean entropy of a marked sequence was 4.473 in Carbon and 1.074 in GENERator. The correlation was measured in the local cohort, which does not include the weakest sequences of the development cohort, so it does not explain them.

5 Discussion

A verifier that knows neither the strand, the start of the generated region, nor the token phase can recover a generation-time watermark from DNA, with one decision corrected for the whole search. The verifier is assembled from existing parts: the SynthID detection statistic, a search over contiguous windows of the kind used for text [25], and a Bonferroni correction. What is specific to DNA is the search space, which adds strands and token phases to start positions and lengths, and the single correction that covers it. This shared watermarking and detection procedure is the methodological contribution. The local comparison measures what the search buys and what it does not (Section 4.8). It recovers detection that an aligned oracle loses after an insertion, and its median corrected strength was higher than that of a phase-only search for short regions in long reads. Where the layout is aligned, it adds no detections to the oracle's.

The verifier tolerates random edits because it does not need an intact window. A simple argument predicts failure much earlier: at a 1% rate, stretches free of insertions and deletions average about 100 bases, shorter than the 384-base minimum window. A window, however, gains excess 1s from every token that keeps its phase and context, and the other tokens only dilute the count. An unedited sequence starts near strength 800 against a threshold of 6.21, so it can lose most of its evidence before the decision changes. The shift toward shorter windows after an insertion or deletion fits the same picture, with the search finding the original phase again downstream of the edit. The strongest window does not identify the exact generation boundaries.

Running the same sampler and verifier in two models shows that the implementation carries across them. The two models agreed in detection, had nearly identical typical strength, and showed no measurable quality loss, although they share much of their training data.

Shorter regions lower the margin, and the local layout study measured by how much (Section 4.7). Strength fell in proportion to the region length, from 797.4 at 3,072 bases to 94.0 at 384 bases in GENERator (791.5 and 94.9 in Carbon), while at least 1,023 of 1,024 marked reads were detected in every layout. A region whose length falls between two window lengths had a median strength close to that of the next shorter window length (216.1 at 1,152 bases against 194.8 at 768 bases in GENERator). Flanks of up to 4,608 natural bases on each side add windows to the correction, and they left the median strength, which is not corrected for the search, at the window lengths unchanged. The margin also varies between sequences. The weakest marked sequences lay well below the median (Supplementary Section S6). In the local cohort, mean entropy correlated with strength (0.581 in GENERator, 0.663 in Carbon), but that is a correlation in a different cohort from the weakest sequences, and not a causal attribution (Section 4.10). The shortest detectable region, which lies below the 384 bases tested, and the trade-off between detection power and distortion across tournament depths remain to be measured in both models.

The search addresses a synchronization problem that coding theory treats directly. Synchronization strings add indexing information to recover from insertions and deletions [16], and pseudorandom codes combine indistinguishability with decoding under specified errors [6]. Davey and MacKay's watermark codes integrate over the accumulated drift with a hidden Markov model, but they presume control of the encoding, which a sampling watermark does not have [10]. Our verifier adds no indexing code to the sequence. It enumerates the offsets, pays for the search through its correction, and carries none of the error-correction guarantees of those codes. Two measurements indicate where a different verifier could gain.

The correction is about 19 times more severe than the effective number of tests (Supplementary Section S3), and across 64 keys the observed fractions were about a tenth of their nominal levels (Section 4.9). The decision also rests on the single strongest window, and one insertion or deletion removes about 45% of the median strength (Section 4.4). A threshold calibrated to the dependence between windows, or a score that combines windows on both sides of a phase shift, could address these, and we evaluated neither. The fine-grid search of the local comparison changes only how densely the same statistic is searched. It raised the median corrected strength after an insertion from 451.2 to 591.8 in Carbon and from 460.9 to 596.3 in GENERator, at about five times the cost, and it added no detections. Synonymous-codon watermarks for coding sequences [41] serve a complementary goal, and combining them with generation-time marks would need its own evaluation.

6 Limitations

This evaluation covers two models, one watermark construction, and one verifier, and it gives no general proof of watermark robustness, biological safety, or secret-key security. Both models use six-base tokens; single-base or variable-length tokens would pose different alignment problems. The three cohorts come from the same 16 chromosome records or a subset of them, and the natural flanks of the layout study come from the same 12 records as the prompts. This limits extrapolation to other genomic sources. Longer sequences would increase the search correction, and the layout study went to flanks of 4,608 bases on each side. The models are not independent, because Carbon's training corpus is built mainly on GENERator-v2's pre-training data, and their agreement does not make them equivalent in any respect.

We compared SynthID with its matched ordinary sampler, not with another watermark construction on the same cohort, such as a biased-logit scheme adapted to six-base tokens or another distribution-preserving scheme. That comparison would confound the construction with the verifier that is our contribution, and the property comparison in Section 2 does not substitute for measured performance on a shared cohort. The comparison of detectors (Section 4.8) covers an aligned oracle, a phase-only search, and a denser fine-grid search, in five settings with 256 prompts per model. The fine-grid search stands in for a span search adapted from text watermarking [25], which we did not run. The detection power that each part of the search adds, and whether it repays its share of the correction and of the search cost (Supplementary Section S4), is therefore measured only in those settings. The baselines are analysis tools, and no verifier setting changed.

The development and detection cohorts share one layout. The generated region is always the last 3,072 bases of a 3,456-base sequence, on the forward strand, a multiple of six bases from the first base, and exactly as long as the longest window. The local cohort varied the layout (Section 4.7): region lengths from 384 to 3,072 bases, including three lengths between window lengths, natural flanks of up to 4,608 bases on each side, six token phases, both strands, five placements, and three random crops. Detection was at its ceiling in every layout, at least 1,023 of 1,024 marked reads, so the study locates no boundary at which detection fails. Regions shorter than 384 bases, which is 64 tokens, were not tested, and we did not measure the shortest region the verifier can detect. The flanks are natural genomic DNA from the 12 records that gave the prompts. Layouts were not combined with the edit-rate series, and the insertion in the detector comparison is one edit at the standard layout. Each layout rests on 512 prompts from one local machine.

The sampler and quality results come from one execution per model on the development cohort. The detection cohort was generated and scored in a separate computing environment with new keyed functions, but by the same authors with the same code, so it is not an independent replication. The local cohort is a further execution of the same kind, with 512 prompts on one machine in bfloat16, and it is not an independent replication either. Its entropy analysis used a separate teacher-forced pass and not distributions recorded during sampling. A check on a few sequences per model found the two to agree up to bfloat16 rounding; it is described with the run and is not part of the evidence. For Carbon, the protocol document retained with the development-cohort detection result does not match the hash recorded in that result, and the original document could not be recovered, so the cause of the mismatch is unknown. The run took its settings from the defaults of the detector code and recorded only a hash of the document. An audit found that eight of the nine settings it governs are recorded in the result's own fields and agree with the reported analysis. The ninth, the 1,024-context repetition history, rests only on the default of the detector code identified by its recorded hash. The mismatch therefore gives no indication that the sequences or the detection statistic differ from what we report, but the execution cannot be checked against the document itself. In the detection cohort, the recorded hash matches the protocol document for both models.

The false-positive evidence is narrow. With 192 prompts, the development cohort could not show a rate below 1% (Supplementary Table S1). In the detection cohort, every bound for ordinary sequences lies below 1%, but the largest, 0.850%, does so narrowly, and the wrong-key bound for Carbon after a deletion exceeds that level at 1.015%. In the local cohort a control needs at most one positive prompt for its bound to fall below 1% at 512 prompts, and several controls with two or more positive prompts have bounds above that level (Section 4.6; Supplementary Table S3). One GENERator layout, the 384-base region, had a prompt-level ordinary rate above 1% (Section 4.7), and the aligned oracle and the phase-only search exceeded 1% at prompt level in several settings (Section 4.8). These rates apply to the sequence length and sequence sources tested here; elsewhere, false-positive control rests on the Bonferroni correction alone.

That correction is valid only if each window P-value is valid, which requires mark bits rebuilt without the correct key to behave as independent fair bits. The aligned-detector diagnostic (Supplementary Section S7) checks this for GENERator with the token phase known. In the detection cohort we checked the full search through its realized rates on ordinary sequences and wrong-key controls (Section 4.3; Figure 5c), scored with the two published keys under one domain label per model and cohort. In the local cohort we checked it under 64 keys never used in generation, on ordinary output, marked output, and 1,608 natural genomic spans (Section 4.9), and no fraction exceeded its nominal level. That evidence is bounded. The 64 keys come from one derivation and one domain label. The model-output reads are at the standard layout and the natural spans are 3,072 bases, so calibration at other sequence lengths is unmeasured. The natural spans come from the same 12 records as the prompts, not from other genomes. Pooled decisions share reads and keys, so the pooled intervals treat dependent decisions as independent.

Every edit was placed at random, without reference to the detector, and the edit-rate series scored no wrong-key controls. Each insertion or deletion spanned at most five bases at a randomly drawn position. We did not test clustered edits, longer insertions or deletions, or recombination of marked and unmarked segments, and the results are not broken down by edit position. Random cropping and embedding in natural flanks were tested in the layout study, but not combined with edits. We did not test an adaptive editor who inspects the detector, the key, or any score and places edits where they do the most damage.

The keys are published for reproducibility, so secrecy, resistance to key recovery, and removal in deployment were not evaluated, and the analytic null assumes independent fair mark bits without the correct key. Attacks reported against text watermarks remain untested here, including key recovery from the detector's P-values [35], paraphrasing and back-translation of SynthID-Text [17], and a layer-inflation attack shown to affect the mean score that our verifier also uses [33].

Model likelihood and sequence statistics are proxies, not measures of expression, replication, function, viability, or safety, and our quality claim covers only the 14 measures. Standardized in-silico suites, such as supervised downstream tasks and zero-shot variant-effect benchmarks [30, 29, 34], score a model's reading of annotated DNA and mostly rely on labels tied to genome coordinates that a new generation does not carry. Evaluating marked outputs against them would need a paired design or an experimental assay, which is the natural next step before any claim about biological consequence.

7 Conclusion

The SynthID tournament watermark can be embedded in Carbon-500M and GENERator-v2 1.2B without a statistically detectable change in the evaluated quality proxies. A verifier holding the key can recover it from DNA by searching strands, start positions, and window lengths under one correction. In the detection cohort, the verifier found every marked sequence in both models, unedited and after one edited base, and the one-sided 95% upper confidence bound on the false-positive rate for ordinary sequences was at most 0.850%. Detection survived random edits up to a 2% per-base rate and failed for most sequences at 10%. A local cohort of 512 prompts, generated on a different machine by the same authors, detected at least 1,023 of 1,024 marked reads in each of 54 layouts per model, and no calibration fraction exceeded its nominal level under 64 unused keys. Broader conclusions require independent replication, regions shorter than 384 bases, other genomic sources, tests against adaptive editing, and biological validation.

AI use statement

Generative AI tools assisted with method implementation, experiment design, statistical derivations, data preparation, literature retrieval, and interpretation of results. They also assisted with drafting and revising the manuscript, articulating the verification setting and claim scope, and preparing the LaTeX submission. Carbon-500M and GENERator-v2 1.2B generated the DNA continuations studied in the experiments, as described in Section 3. The paper contains no formal theorems or proofs. Its statistical development uses standard probability models and a Bonferroni correction, and the authors verified all derivations, numbers, and claims against the recorded measurements. The authors are responsible for the final text, claims, and artifacts.

Ethics statement

The prompt cohorts use public non-human RefSeq records and exclude patient and controlled-access sequences (Section 3.4). The experiments assess statistical detection and sequence proxies. They do not establish biological function, viability, safety, or sequence authenticity. The experimental keys are published for reproducibility and provide no evidence of secrecy or resistance to key recovery. A detected watermark is evidence under the specified statistical model, not a certificate of biological origin or safe use.

Reproducibility statement

Section 3 specifies the canonical token policy, the tournament update, the detector search and its correction, prompt construction, and the statistical comparisons. Section 3.9 specifies the local cohort: its layouts, the baseline detectors, the unused keys, and the entropy pass.

The Supplementary information gives the window-score selection, the search cost, and the computing environments, including the local machine. The figures and tables report both models separately under the same evaluation design. Section 6 records the execution scope, the absence of an independent replication, and the Carbon protocol-document discrepancy together with its audit. The code, protocol documents, and evidence ledger needed to reproduce the analysis are available as described in the Data and code availability statement.

Data and code availability

The verifier implementation, the watermark sampler, the prompt-cohort construction scripts, the analysis and figure scripts, the protocol documents, and the evidence ledger to which every number in this paper resolves are maintained in the project GitHub repository. The repository includes the detection results for both models in the detection cohort, the GENERator results for the development cohort, and the protocol, execution record, and compact result files of the local cohort for both models. The per-read trial files and generated sequences of the local cohort are available from the authors on request. The Carbon development-cohort result files, generated sequences, and other underlying files not in the repository are available from the authors on request. The sampler's implementation tests, including the comparison with the SynthID-Text reference implementation at the cited commit, are part of the repository's test suite. The two experimental keys are published for reproducibility and are not deployment secrets.

Competing interests

The authors declare no competing financial or non-financial interests.

Acknowledgments

This work has been supported by the National Institute of General Medical Sciences of the National Institutes of Health [R35GM155468 to I.G.S.]; and start-up funds awarded to I.G.S.

References

[1] Scott Aaronson and Hendrik Kirchner. Watermarking GPT outputs. Talk slides, 2022. https://www.scottaaronson.com/talks/watermark.ppt.

[2] Masanori Arita and Yoshiaki Ohashi. Secret signatures inside genomic DNA. *Biotechnology Progress*, 20(5):1605–1607, 2004.

[3] David Baker and George Church. Protein design meets biosecurity. *Science*, 383(6681):349, 2024. Editorial.

[4] Loubna Ben Allal, Qiuyi Li, Maurizio Fiusco, Lewis Tunstall, Kashif Rasul, Ed Beeching, Dana Aubakirova, Carlos Patiño, Thibaud Frere, Anton Lozhkov, Georgia Channing, Thomas Wolf, Diego di Bernardo, and Leandro von Werra. Carbon: Decoding the language of life. *bioRxiv*, 2026. Preprint, version 1, posted 25 May 2026, doi:10.64898/2026.05.22.727119.

[5] Yanshuo Chen, Zhengmian Hu, Yihan Wu, Ruibo Chen, Yongrui Jin, Marcus Zhan, Chengjin Xie, Wei Chen, and Heng Huang. Enhancing privacy in biosecurity with watermarked protein design. *Bioinformatics*, 41(7):btaf141, 2025.

[6] Miranda Christ and Sam Gunn. Pseudorandom error-correcting codes. In *Advances in Cryptology — CRYPTO 2024*, Lecture Notes in Computer Science, pages 325–347. Springer, 2024. IACR ePrint 2024/235.

[7] Miranda Christ, Sam Gunn, and Or Zamir. Undetectable watermarks for language models. In *Proceedings of the 37th Annual Conference on Learning Theory (COLT)*, volume 247 of *PMLR*, pages 1125–1139, 2024. arXiv:2306.09194.

[8] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, et al. Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025.

[9] Sumanth Dathathri, Abigail See, Sumedh Ghaisas, Po-Sen Huang, Rob McAdam, Johannes Welbl, Vandana Bachani, Alex Kaskasoli, Robert Stanforth, Tatiana Matejovicova, Jamie Hayes, Nidhi Vyas, Majd Al Merey, Jonah Brown-Cohen, Rudy Bunel, Borja Balle, Taylan Cemgil, Zahra Ahmed, Kitty Stacpoole, Ilia Shumailov, Ciprian Baetu, Sven Gowal, Demis Hassabis, and Pushmeet Kohli. Scalable watermarking for identifying large language model outputs. *Nature*, 634(8035):818–823, 2024.

[10] Matthew C. Davey and David J. C. MacKay. Reliable communication over channels with insertions, deletions, and substitutions. *IEEE Transactions on Information Theory*, 47(2):687–698, 2001.

[11] Abdulrahman Diaa, Toluwani Aremu, and Nils Lukas. Optimizing adaptive attacks against watermarks for language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 13546–13569, 2025.

[12] GenerTeam. GENERator-v2-eukaryote-1.2b-base model card and released modeling code. https://huggingface.co/GenerTeam/GENERator-v2-eukaryote-1.2b-base, 2026.

[13] Daniel G. Gibson, John I. Glass, Carole Lartigue, Vladimir N. Noskov, Ray-Yuan Chuang, Mikkel A. Algire, Gwynedd A. Benders, Michael G. Montague, Li Ma, et al. Creation of a bacterial cell controlled by a chemically synthesized genome. *Science*, 329(5987):52–56, 2010.

[14] Noah Golowich and Ankur Moitra. Edit distance robust watermarks via indexing pseudorandom codes. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, volume 37, pages 20645–20693. Curran Associates, Inc., 2024. arXiv:2406.02633.

[15] Google DeepMind. SynthID-Text reference implementation. https://github.com/google-deepmind/synthid-text, 2024. Commit addb4a15, 13 June 2025.

[16] Bernhard Haeupler and Amirbehshad Shahrasbi. Synchronization strings: Codes for insertions and deletions approaching the Singleton bound. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, pages 33–46, 2017.

[17] Xia Han, Qi Li, Jianbing Ni, and Mohammad Zulkernine. Robustness assessment and enhancement of text watermarking for Google's SynthID, 2025. arXiv:2508.20228; accepted at IEEE TrustCom 2025.

[18] Dominik Heider and Angelika Barnekow. DNA-based watermarks using the DNA-Crypt algorithm. *BMC Bioinformatics*, 8:176, 2007.

[19] Abe Hou, Jingyu Zhang, Tianxing He, Yichen Wang, Yung-Sung Chuang, Hongwei Wang, Lingfeng Shen, Benjamin Van Durme, Daniel Khashabi, and Yulia Tsvetkov. SemStamp: A semantic watermark with paraphrastic robustness for text generation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4067–4082. Association for Computational Linguistics, 2024.

[20] Zhengmian Hu, Lichang Chen, Xidong Wu, Yihan Wu, Hongyang Zhang, and Heng Huang. Unbiased watermark for large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.10669.

[21] HuggingFaceBio. Carbon-500M model card. https://huggingface.co/HuggingFaceBio/Carbon-500M, 2026.

[22] Nikola Jovanović, Robin Staab, and Martin Vechev. Watermark stealing in large language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 22570–22593, 2024. arXiv:2402.19361.

[23] Samuel H. King, Claudia L. Driscoll, David B. Li, Daniel Guo, Aditi T. Merchant, Garyk Brixi, Max E. Wilkinson, and Brian L. Hie. Generative design of bacteriophages with genome language models. *Science*, 393(6811):eaec2657, 2026. PMID: 42561074.

[24] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 17061–17084, 2023. arXiv:2301.10226.

[25] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Manli Shu, Khalid Saifullah, Kezhi Kong, Kasun Fernando, Aniruddha Saha, Micah Goldblum, and Tom Goldstein. On the reliability of watermarks for large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2306.04634.

[26] Rohith Kuditipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. Robust distortion-free watermarks for language models. *Transactions on Machine Learning Research*, 2024. arXiv:2307.15593.

[27] Qiuyi Li, Zhihao Zhan, Shikun Feng, Yiheng Zhu, Yuan He, Wei Wu, Zhenghang Shi, Shengjie Wang, Zongyong Hu, Zhao Yang, Jiaoyang Li, Jian Tang, Haiguang Liu, and Tao Qin. GENERator-v2: Reconciling coarse tokenization with single-nucleotide resolution in genomic language modeling. *bioRxiv*, 2026. Preprint, version 2, posted 4 May 2026, doi:10.64898/2026.01.27.702015.

[28] Michael Liss, Daniela Daubert, Kathrin Brunner, Kristina Kliche, Ulrich Hammes, Andreas Leiherer, and Ralf Wagner. Embedding permanent watermarks in synthetic genes. *PLoS ONE*, 7(8):e42465, 2012.

[29] Zicheng Liu, Jiahui Li, Siyuan Li, Zelin Zang, Cheng Tan, Yufei Huang, Yajing Bai, and Stan Z. Li. GenBench: A benchmarking suite for systematic evaluation of genomic foundation models, 2024. arXiv:2406.01627.

[30] Frederikke Isa Marin, Felix Teufel, Marc Horlacher, Dennis Madsen, Dennis Pultz, Ole Winther, and Wouter Boomsma. BEND: Benchmarking DNA language models on biologically meaningful tasks. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2311.12570.

[31] Eric Nguyen, Michael Poli, Matthew G. Durrant, Brian Kang, Dhruva Katrekar, David B. Li, Liam J. Bartie, Armin W. Thomas, Samuel H. King, Garyk Brixi, et al. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723):eado9336, 2024.

[32] Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Michael Wornow, Callum Birch-Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, Stefano Ermon, Christopher Ré, and Stephen Baccus. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, pages 43177–43201, 2023. arXiv:2306.15794.

[33] Romina Omidi, Yun Dong, and Binghui Wang. On Google's SynthID-Text LLM watermarking system: Theoretical analysis and empirical validation, 2026. arXiv:2603.03410.

[34] Aman Patel, Arpita Singhal, Austin Wang, Anusri Pampari, Maya Kasowski, and Anshul Kundaje. DART-Eval: A comprehensive DNA language model evaluation benchmark on regulatory DNA. In *Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track*, volume 37, pages 62024–62061, 2024. arXiv:2412.05430.

[35] Shayleen Reynolds, Hengzhi He, Dung Daniel T. Ngo, Saheed Obitayo, Niccolò Dalmasso, Guang Cheng, Vamsi K. Potluru, and Manuela Veloso. Breaking distortion-free watermarks in large language models, 2025.

[36] Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. Can AI-generated text be reliably detected? stress testing AI text detectors under various attacks. *Transactions on Machine Learning Research*, 2025.

[37] Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range DNA sequence modeling. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 43632–43648, 2024. arXiv:2403.03234.

[38] Wei Wu, Qiuyi Li, Yuanyuan Zhang, Zhihao Zhan, Ruipu Chen, Mingyang Li, Kun Fu, Junyan Qi, Yongzhou Bao, Chao Wang, et al. GENERator: A long-context generative genomic foundation model. *arXiv preprint arXiv:2502.07272*, 2025. Version 5, 31 January 2026.

[39] Yihan Wu, Zhengmian Hu, Junfeng Guo, Hongyang Zhang, and Heng Huang. A resilient and accessible distribution-preserving watermark for large language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 53443–53470, 2024. arXiv:2310.07710.

[40] Hanlin Zhang, Benjamin L. Edelman, Danilo Francati, Daniele Venturi, Giuseppe Ateniese, and Boaz Barak. Watermarks in the sand: Impossibility of strong watermarking for language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 58851–58880, 2024. arXiv:2311.04378.

[41] Zaixi Zhang, Ruofan Jin, Le Cong, and Mengdi Wang. Securing the language of life: Inheritable watermarks from DNA language models to proteins. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*, volume 38, pages 152050–152080, 2025. arXiv:2509.18207.

[42] Xuandong Zhao, Prabhanjan Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for AI-generated text. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2306.17439.

Supplementary information

S1 Window-score selection

Before any detection result was examined, 64 development prompts were reserved for choosing how to summarize the mark bits in a window. Neither they nor the 192 held-out prompts were used to fit the detection threshold. The two candidates were the released SynthID mean score over scored tokens and the upstream default weighted-mean score. For GENERator, the two were compared on the 64 reserved prompts alone. The comparison used continuations of 384 and 3,072 bases, scored with the token phase known, and measured the separation between marked and ordinary continuations in units of the ordinary-continuation standard deviation. The mean score separated them at least as well at both lengths and was selected. No held-out prompt was scored during this selection, and no signal-matched weights were fitted. No other verifier setting was fitted to the reserved prompts: the window lengths and the significance level are fixed settings of the verifier, and the threshold follows from them and the sequence length. We applied the same mean score to Carbon, but the equivalent Carbon comparison is recorded only in the run notes, not as a retained result, so this selection is independently verified for GENERator only.

S2 Window geometry

Each scored token carries 30 mark bits, one per tournament layer. The number of bits does not limit detection: 21 bits that are all 1 already reach the corrected threshold, so one scored token would suffice. What limits detection is dilution. The shortest window, 384 bases, holds 64 tokens. The first four supply unwatermarked context, which leaves at most 60 scored tokens and at most 1,800 mark bits, fewer where a four-token context repeats. With all 60 scored, at least 1,004 of the 1,800 bits must be 1, about 56%. Longer windows need a smaller fraction. Detection therefore depends on recoverable synchronization, not on the absence of edits.

S3 Conservatism of the correction

Bonferroni correction bounds the family-wise error rate without requiring independence between tests. Because the searched windows are positively correlated, the effective number of independent tests is smaller than M. The realized false-positive rate can then lie well below α, and recall is attained under a stricter threshold than necessary. The realized per-sequence rates in the detection cohort (Section 4.3) are 9.5 times below α for Carbon and 5.4 times below for GENERator. The observed minimum window P-values behave like those of about 828–852 independent tests, consistently across both models and all four conditions. That is about 19 times fewer than the 16,136 windows searched. This moment-matched figure summarizes how conservative the correction is. It does not replace the decision rule, which is unchanged.

S4 Search cost and computing environment

Keyed hashing, not window enumeration, dominates the cost of the search. A sequence of N bases has twelve token frames, two orientations by six phases, and the mark bits of every token position in a frame are computed once, in a single pass. Window statistics then follow from prefix sums over those per-token bit counts, so each window of a given length is evaluated in constant time. Where a frame contains repeated four-token contexts, the first-occurrence rule is enforced with Fenwick trees, and each window costs $O(\log N)$ instead. The work is therefore $O(N)$ keyed hash evaluations and $O(N \log N)$ additional time, with $O(N)$ space per sequence, and M grows linearly in N for a fixed window set. Sequences shorter than every window length are not scored, reverse-complement coordinates are mapped back onto the supplied sequence, and log-probabilities are kept throughout to avoid floating-point underflow.

Timing was measured on an AMD EPYC 7763. One 3,456-base sequence took a median 0.0956 s on a single thread. Sequences of 6,912 bases took 0.245 s, and sequences of 13,824 bases took 0.902 s. Peak memory per process was 27.9 MiB for a 3,456-base sequence. For a 13,824-base sequence, it was 69.1 MiB. Running 16 workers in parallel across sequences gave one sequence per 0.0167 s of wall-clock time. That is 5.7 times the single-threaded rate.

Development-cohort generation ran on GPUs. Model scoring ran on CPU for Carbon and on GPU for GENERator. The detection cohort was generated on NVIDIA A100 GPUs in a separate computing environment. Detection ran on x86-64 CPUs for both cohorts.

The local cohort was generated, and its entropy pass run, on one Apple M5 Pro machine with 18 CPU cores and 48 GiB of unified memory, in bfloat16 on the PyTorch 2.13.0 MPS backend with Python 3.12. The model revisions were those of the other cohorts, loaded from a local cache without network access. Three generation processes per model shared the device, and generation produced 193.1 bases per summed process-second in GENERator and 88.2 in Carbon. Detection, calibration, and the single-thread timing ran on the same machine's CPU, with 12 worker processes except for timing, which used one process. The verifier took a median 44.4 ms per read in GENERator and 45.1 ms in Carbon at the standard layout (Table 3), against 95.6 ms on the AMD EPYC processor above, so the two timings are not comparable across processors.

S5 Development-cohort detection

Table S1 gives the development-cohort counts, and Figures S1 and S2 show strength and strongest-window length. Every marked sequence was found in both models and every condition. Each control produced at most one positive sequence per model and condition, and the GENERator wrong-key positive is absent after a deletion. With 192 prompts, the exact control intervals reach 2.868% for one positive prompt and 1.903% for none. This cohort therefore cannot show whether the false-positive rate lies below 1%; the detection cohort addresses that question.

S6 Strength and the weakest sequence

Typical strength is nearly the same in both models. In the development cohort, the median unedited marked sequence had strength 790.6 in Carbon and 798.8 in GENERator. Control strengths were concentrated near or below the threshold, and the few positive controls lay only slightly above it (Figure S1a,c). The weakest marked sequence differs more between the models. In the development cohort, it had strength 124.8 in Carbon and 19.6 in GENERator. In the detection cohort, the weakest sequences were 237.6 in Carbon and 123.8 in GENERator, and the medians were again close (Table S2). The difference between weakest sequences therefore reflects single sequences, not the models' typical margin.

The weakest detected GENERator sequence in the development cohort contained a highly repetitive continuation. Of its four-token contexts, 174 repeated and were excluded from scoring, which left 334 of a possible 508 tokens. Only 54.6% of its mark bits were 1, against a median of 74% in both models. Restoring all 508 scored tokens at that fraction would close only 9% of the gap to Carbon's weakest sequence, so most of the shortfall comes from weak signal in the tokens that remained. The tournament can raise the fraction of 1s only where the model spreads probability across several tokens, which is consistent with a low-entropy, repetitive stretch carrying little signal. We did not measure per-token entropy for this sequence. In the local cohort, which does not include it, mean entropy and strength were positively correlated (Section 4.10). We therefore find no evidence that typical detection strength differs between the models.

S7 Aligned-detector diagnostic in GENERator

For GENERator, a separate diagnostic checked the analytic null with a detector given the known token phase. Across the four evaluated lengths, 2, 6, 6, and 4 ordinary prompts were positive, against about five expected at each length ($P = 0.25$–0.87). This result is compatible with the analytic null, but it does not replace the full-search correction used for the position-independent decision, and it is not a calibration of the full search, which Section 4.9 addresses across 64 keys in the local cohort.

S8 Strength and window length by edit

Table S2 gives the median strength of marked sequences in the detection cohort and, for both cohorts, how often the full 3,072-base window carried the strongest evidence.

S9 Local cohort: standard layout

Table S3 gives the standard-layout results of the local cohort in the format of Table 1. The two tables are not pooled.

S10 Layout study: all layouts

Table S4 lists every layout of the layout study for both models.

S11 Detector comparison: prompt-level counts

Table S5 gives the prompt-level counts of the detector comparison. A prompt is positive for a marked read when both draws are detected, and positive for an ordinary read when either draw is, so the prompt-level control rate is a union over two reads.

S12 Calibration across keys: all levels

Table S6 gives the fraction of decisions whose sequence-level P-value is at or below each nominal level, with the count in parentheses, for 32,768 decisions per model and kind of output and 102,912 for natural spans. No fraction exceeds its nominal level, and no one-sided 95% lower bound exceeds its nominal level.

Table S1: Positive detections in the development cohort, for each model across 192 held-out prompts with two draws each. Correct-key and wrong-key columns refer to marked sequences. Each edit changes exactly one base. The final two rows summarize unedited sequences at the prompt level to account for paired draws; intervals are exact and two-sided. All marked prompts also had both draws detected in every edit condition.

	Carbon-500M			**GENERator-v2 1.2B**		
Edit applied to the sequence	Correct key	Ordinary	Wrong key	Correct key	Ordinary	Wrong key
None	384/384	1/384	0/384	384/384	0/384	1/384
One substitution	384/384	1/384	0/384	384/384	0/384	1/384
One insertion	384/384	1/384	0/384	384/384	0/384	1/384
One deletion	384/384	1/384	0/384	384/384	0/384	0/384
Prompts with a positive draw	192/192	1/192	0/192	192/192	0/192	1/192
Exact 95% interval	98.10–100%	0.013–2.868%	0–1.903%	98.10–100%	0–1.903%	0.013–2.868%

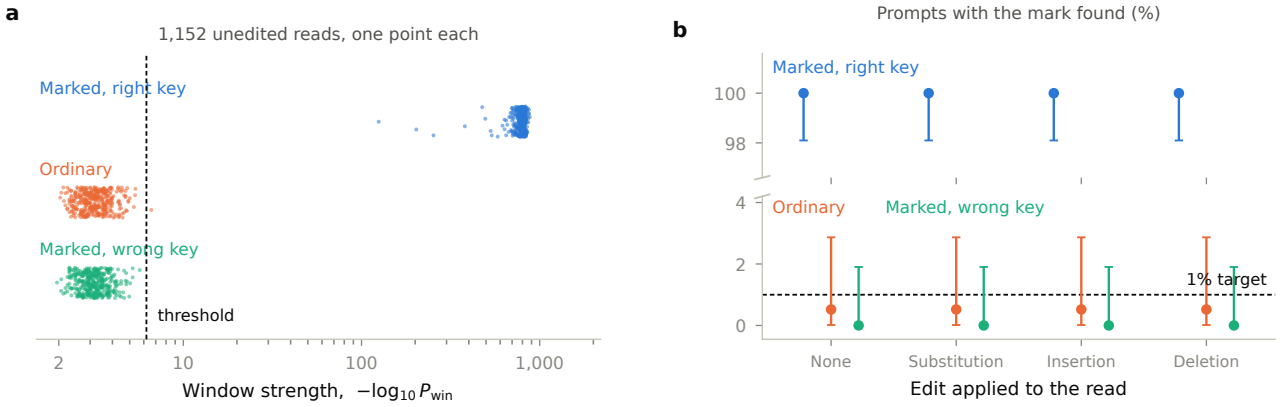
Table S2: Strength and strongest-window length of marked sequences, by edit. Median strength is for the detection cohort, over 3,088 sequences per model and condition. The full-window columns count sequences whose strongest window was the full 3,072-base window, out of 384 per model and condition in the development cohort and 3,088 in the detection cohort. Each edit changes exactly one base.

	Median strength		**Full window, development**		**Full window, detection**	
Edit applied to the sequence	Carbon	GENERator	Carbon	GENERator	Carbon	GENERator
None	792.9	799.7	383	381	3,087	3,085
One substitution	777.6	783.8	383	381	3,087	3,085
One insertion	441.8	443.1	203	200	1,724	1,732
One deletion	437.7	438.5	232	227	1,663	1,671

Table S3: Detection at the standard layout in the local cohort, for each model across 512 prompts with two draws each. *Correct key* counts marked reads detected (1,024 per model and condition), and every marked prompt had both draws detected (512 of 512). *Strength* is the median strength of marked reads. *Ordinary* and *Wrong key* count positive prompts (a prompt is positive if either draw is); the value in parentheses is the exact one-sided 95% upper confidence bound on the prompt-level rate, in percent. Each edit changes exactly one base.

	Carbon-500M				**GENERator-v2 1.2B**			
Edit applied to the sequence	Correct key	Strength	Ordinary	Wrong key	Correct key	Strength	Ordinary	Wrong key
None	1,024/1,024	791.5	3 (1.507)	2 (1.224)	1,024/1,024	797.4	0 (0.583)	2 (1.224)
One substitution	1,024/1,024	775.8	2 (1.224)	2 (1.224)	1,024/1,024	780.7	0 (0.583)	2 (1.224)
One insertion	1,024/1,024	444.3	2 (1.224)	1 (0.923)	1,024/1,024	450.9	0 (0.583)	2 (1.224)
One deletion	1,024/1,024	437.4	2 (1.224)	1 (0.923)	1,024/1,024	437.5	1 (0.923)	1 (0.923)

Carbon-500M



GENERator-v2 1.2B

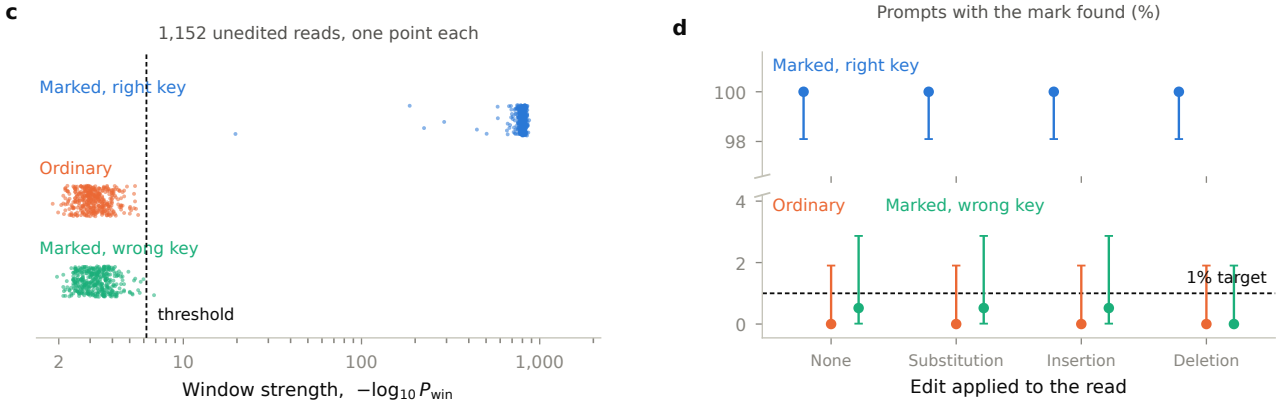


Figure S1: Position-independent detection in the development cohort. **a,c**, Strength of the best window in each unedited sequence, one point per sequence, on a log scale; in the panel labels, "Marked, right key" denotes marked sequences scored with the correct key, and "reads" denotes test sequences. The dashed line is the threshold corrected for the search over 16,136 windows. All marked sequences exceed the threshold. **b,d**, Proportion of the 192 held-out prompts with positive detections, with exact two-sided 95% intervals, on a broken axis. All marked prompts were detected in all four edit conditions. The GENERator wrong-key positive is absent after a deletion. With 192 prompts, the control intervals extend above the 1% target; Table 1 and Figure 3 report the detection cohort. Panels **a,b** show Carbon and panels **c,d** show GENERator.

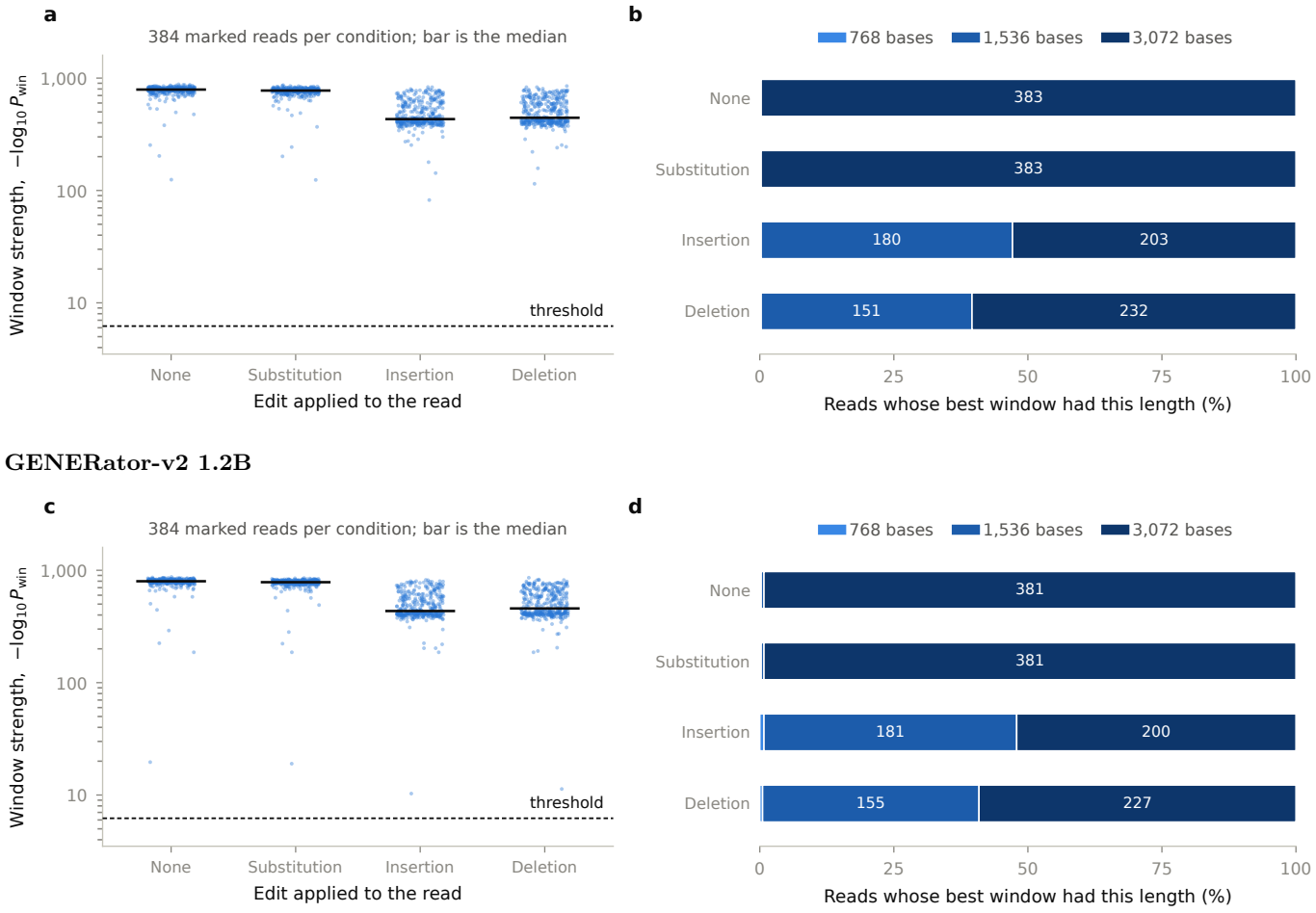


Figure S2: Single-base edits in the development cohort. In the panel labels, "reads" denotes test sequences. **a,c**, Strength of the best window in each of the 384 marked sequences per condition, on a log scale, with the median shown as a bar. The dashed line is the detection threshold. **b,d**, Length of the strongest window; numbers inside the bars are sequence counts out of 384. As in the detection cohort (Figure 4), the full 3,072-base window is strongest for nearly all unedited and substituted sequences, whereas shorter windows are more frequent after an insertion or deletion. Panels **a,b** show Carbon and panels **c,d** show GENERator.

Table S4: All 54 layouts of the layout study, local cohort. *Reads* are marked reads detected (of 1,024), *Prompts* are prompts with both draws detected (of 512), and *Strength* is the median strength of the best window before the correction for the search. *Ord. reads* are ordinary reads flagged (of 1,024), and *Ord. prompts* are prompts positive on either draw (of 512), with the exact one-sided 95% upper confidence bound on that prompt rate in parentheses, in percent. Series A: the region follows the prompt, length in bases. B: the region between natural flanks in a 6,144-base read. C: a region of 768 or 3,072 bases after a left flank of $384 + o$ bases, with phase o, supplied forward (fwd) or as its reverse complement (rev). D: region length plus the flank on each side. E: a 768-base region starting at the given base of a 6,144-base read. F: a random crop of the standard read of the given length.

	Carbon-500M					**GENERator-v2 1.2B**				
Layout	Reads	Prompts	Strength	Ord. reads	Ord. prompts (bound)	Reads	Prompts	Strength	Ord. reads	Ord. prompts (bound)
A, 384	1,024	512	94.9	3	3 (1.51)	1,023	511	94.0	6	6 (2.30)
A, 576	1,024	512	106.8	3	3 (1.51)	1,023	511	106.5	5	5 (2.04)
A, 768	1,024	512	194.6	5	5 (2.04)	1,023	511	194.8	3	3 (1.51)
A, 1,152	1,024	512	216.5	5	5 (2.04)	1,024	512	216.1	3	3 (1.51)
A, 1,536	1,024	512	394.1	4	4 (1.78)	1,024	512	394.5	1	1 (0.92)
A, 2,304	1,024	512	408.7	3	3 (1.51)	1,024	512	409.9	1	1 (0.92)
A, 3,072	1,024	512	791.5	3	3 (1.51)	1,024	512	797.4	0	0 (0.58)
B, 384	1,023	511	94.9	3	3 (1.51)	1,023	511	94.0	0	0 (0.58)
B, 576	1,024	512	110.8	2	2 (1.22)	1,023	511	110.4	3	3 (1.51)
B, 768	1,024	512	194.4	0	0 (0.58)	1,023	511	194.8	2	2 (1.22)
B, 1,152	1,024	512	223.4	1	1 (0.92)	1,024	512	223.7	2	2 (1.22)
B, 1,536	1,024	512	394.1	0	0 (0.58)	1,024	512	394.5	1	1 (0.92)
B, 2,304	1,024	512	446.7	1	1 (0.92)	1,024	512	448.7	1	1 (0.92)
B, 3,072	1,024	512	791.5	1	1 (0.92)	1,024	512	797.4	1	1 (0.92)
C, 768, phase 0, fwd	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	1	1 (0.92)
C, 768, phase 0, rev	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	1	1 (0.92)
C, 768, phase 1, fwd	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 1, rev	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 2, fwd	1,024	512	194.6	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 2, rev	1,024	512	194.6	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 3, fwd	1,024	512	194.8	2	2 (1.22)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 3, rev	1,024	512	194.8	2	2 (1.22)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 4, fwd	1,024	512	194.6	5	5 (2.04)	1,023	511	194.8	1	1 (0.92)
C, 768, phase 4, rev	1,024	512	194.6	5	5 (2.04)	1,023	511	194.8	1	1 (0.92)
C, 768, phase 5, fwd	1,024	512	194.6	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 768, phase 5, rev	1,024	512	194.6	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
C, 3,072, phase 0, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	1	1 (0.92)
C, 3,072, phase 0, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	1	1 (0.92)
C, 3,072, phase 1, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 1, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 2, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 2, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 3, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	1	1 (0.92)
C, 3,072, phase 3, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	1	1 (0.92)
C, 3,072, phase 4, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 4, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 5, fwd	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
C, 3,072, phase 5, rev	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
D, 768 + 0	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	3	3 (1.51)
D, 768 + 384	1,024	512	194.4	4	4 (1.78)	1,023	511	194.8	2	2 (1.22)
D, 768 + 1,536	1,024	512	194.8	3	3 (1.51)	1,023	511	194.8	1	1 (0.92)
D, 768 + 4,608	1,024	512	194.6	1	1 (0.92)	1,023	511	194.8	0	0 (0.58)
D, 3,072 + 0	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
D, 3,072 + 384	1,024	512	791.5	2	2 (1.22)	1,024	512	797.4	0	0 (0.58)
D, 3,072 + 1,536	1,024	512	791.5	0	0 (0.58)	1,024	512	797.4	2	2 (1.22)
D, 3,072 + 4,608	1,024	512	791.5	1	1 (0.92)	1,024	512	797.4	1	1 (0.92)
E, start 0	1,024	512	194.6	1	1 (0.92)	1,023	511	194.8	1	1 (0.92)
E, start 1,344	1,024	512	194.6	3	3 (1.51)	1,023	511	194.8	0	0 (0.58)
E, start 2,688	1,024	512	194.6	2	2 (1.22)	1,023	511	194.8	1	1 (0.92)
E, start 4,032	1,024	512	194.6	4	4 (1.78)	1,023	511	194.8	1	1 (0.92)
E, start 5,376	1,024	512	194.6	0	0 (0.58)	1,023	511	194.8	0	0 (0.58)
F, crop 768	1,024	512	105.2	3	3 (1.51)	1,023	511	105.7	5	5 (2.04)
F, crop 1,536	1,024	512	212.5	3	3 (1.51)	1,024	512	211.1	2	2 (1.22)
F, crop 2,304	1,024	512	407.8	3	3 (1.51)	1,024	512	409.7	0	0 (0.58)

Table S5: Prompt-level counts for the detector comparison in the local cohort, 256 prompts per model. *Marked prompts* are prompts with both draws detected (of 256). *Ordinary prompts* are prompts with a positive ordinary read on either draw, with the exact one-sided 95% upper confidence bound in parentheses, in percent.

	Carbon-500M		**GENERator-v2 1.2B**	
Detector	Marked prompts	Ordinary prompts (bound)	Marked prompts	Ordinary prompts (bound)
Standard layout				
Aligned oracle	256	2 (2.44)	256	4 (3.54)
Phase only	256	6 (4.57)	256	5 (4.06)
Verifier	256	2 (2.44)	256	0 (1.16)
Fine grid	256	0 (1.16)	256	0 (1.16)
Standard layout, one insertion				
Aligned oracle	230	3 (3.00)	227	2 (2.44)
Phase only	256	3 (3.00)	256	2 (2.44)
Verifier	256	1 (1.84)	256	0 (1.16)
Fine grid	256	0 (1.16)	256	0 (1.16)
768-base region, embedded				
Aligned oracle	256	5 (4.06)	256	1 (1.84)
Phase only	255	2 (2.44)	255	5 (4.06)
Verifier	256	0 (1.16)	256	1 (1.84)
Fine grid	256	0 (1.16)	256	0 (1.16)
1,152-base region, embedded				
Aligned oracle	256	4 (3.54)	256	2 (2.44)
Phase only	255	5 (4.06)	256	7 (5.07)
Verifier	256	1 (1.84)	256	2 (2.44)
Fine grid	256	0 (1.16)	256	1 (1.84)
3,072-base region, embedded				
Aligned oracle	256	2 (2.44)	256	4 (3.54)
Phase only	256	2 (2.44)	256	4 (3.54)
Verifier	256	1 (1.84)	256	0 (1.16)
Fine grid	256	1 (1.84)	256	0 (1.16)

Table S6: Calibration of the sequence-level P-value under 64 keys never used in generation, local cohort. Entries are the percentage of decisions with a corrected P-value at or below the nominal level, with the count in parentheses. The decisions share reads and keys.

	Nominal level (%)				
Source of reads	1	5	10	25	50
Carbon-500M, ordinary output	0.113 (37)	0.580 (190)	0.983 (322)	2.591 (849)	4.782 (1,567)
GENERator-v2 1.2B, ordinary output	0.110 (36)	0.525 (172)	1.071 (351)	2.585 (847)	4.736 (1,552)
Carbon-500M, marked output	0.140 (46)	0.558 (183)	1.068 (350)	2.521 (826)	4.587 (1,503)
GENERator-v2 1.2B, marked output	0.128 (42)	0.638 (209)	1.147 (376)	2.719 (891)	4.776 (1,565)
Natural genomic spans	0.128 (132)	0.593 (610)	1.122 (1,155)	2.682 (2,760)	4.728 (4,866)